# Frame transport for framed formal monodromy: the lemma, its hypotheses, and the receiver problem

Fifth version, closing the coalesced case

Memo for a reader of *Irrationality of Cubic Threefolds after One Stabilization*

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**Abstract**

Proposition 4.7 of the draft transports framed formal monodromy across the comparison isomorphisms of Iritani and Iritani–Koto, and the transport step itself is asserted in one sentence. This memo states and proves the missing lemma over a differential field, with its hypotheses made explicit; shows that those hypotheses are jointly unsatisfiable over the coefficient receiver the draft builds, so that the lemma as proved has no verified instance in the application; corrects the receiver; and reduces the remaining question to one inequality that depends on a normalization the draft is free to choose. The inequality fails under the draft's current instruction to take the separating weight parameter large, and holds at the cubic endpoint under the smallest admissible choice. Section 4 examines an iterated receiver that would satisfy the inequality trivially, checks it against the sources, and finds that it needs a semipositivity hypothesis on the ambient variety which weak factorization does not supply.

The second version of this memo proposed the framing theory of Hinault, Yu, Zhang and Zhang as the probable repair, on the ground that its decomposition gauge is regular in the loop coordinate. Three compatibility checks against the pinned sources have since been run, and the route does not survive them; Section 5 now reports what those checks found. Two consequences are worth more than the route was. First, no receiver is needed for the comparison isomorphisms at all: Iritani's $\Psi$ and Iritani–Koto's $\Phi$ are polynomial in the loop coordinate with inverses in the same ring, so the transport theorem below applies to them with $\Gamma = 0$. Second, the diagnosis of the second version was wrong — the pro-Laurent bulk gauge is not the draft's own choice but Iritani–Koto's own fundamental solution, which the sources meet and handle by Birkhoff factorization. What is left is a single statement about one bulk displacement, recorded in Section 5. Section 6 then attacks that statement directly through the flatness identities. It proves that multiplicity-one blocks carry no framed monodromy, which is unconditional and leaves only coalesced blocks in play, and writes down the exact evolution of the block's leading operator. It does *not* reduce the coalesced case: the third version of this memo claimed such a reduction and was wrong twice over, once in dropping a term from the off-block Sylvester equation and once in modelling a nonsemisimple block by its compressed pair. Both errors, and the cubic $2 \times 2$ example that is the regression test for any repair, are recorded there.

Section 7, new in this version, closes the coalesced case for a block of the cubic's type, and with it the residual gap for the cubic ambient summand. The device is to shear the block before deforming it. Three statements are proved over the formal even bulk germ, from the flatness identities and nothing else: the double eigenvalue never splits, because $\det N$ satisfies $\partial_a \det N = 2q_a \det N$ and vanishes initially; no irregularity appears, because the commutant of a regular block forbids a $z^{-2}$ bulk term and the $z^{-1}$ flatness equation then forces the surviving pole to vanish; and the residue of the sheared block moves by conjugation alone, so

its characteristic polynomial is constant. For the cubic that polynomial is $\rho^2+\rho+5/36$ with roots $-1/6, -5/6$, so the primitive-sixth multiplicity is 2 at every bulk parameter, including a specialized one. No receiver, no bulk gauge, and no convergence question enters, because specialization is substitution into constants.

Section 8 assembles the one-stabilization statement — that $X \times \mathbf{P}^1$ is irrational for every smooth cubic threefold — in the atom formulation, where the intermediate varieties of a weak factorization cancel in the ledger instead of being visited, so that no framed operator is transported across a comparison isomorphism and Hypothesis 4.7H does not arise. Every step is proved there except one, and that one is stated as the gap it is: the primitive-sixth count must be invariant under change of base point within a connected component of the spectral cover, which is part of the definition of an atom. Cai's gauge argument gives that invariance along a formal germ, and it does not globalize, because the gauge is the pro-Laurent object whose sum over bulk degrees loses its lower bound in the loop coordinate outside topologically nilpotent parameters. So the receiver difficulty of the earlier sections reappears at the globalization step rather than being escaped. Two consequences are recorded: the low-dimensional side needs no such invariance, because the sources' nef-canonical claim is pointwise structural; and the sources' own architecture avoids the gap by decorating with a Serre automorphism, whose well-definedness on atoms follows from rigidity of representations of a proreductive group rather than from any constancy of monodromy. Section 9 carries the endpoint most of the way. An earlier version of it claimed the theorem outright by inverting the comparison's change of variables; that was refuted, since the sources' invertibility is in displaced coordinates treated as formal variables and the substitution the argument needed is ill-defined. What survives the refutation is stronger than what motivated it. For the trivial rank-two bundle the displacement is $\varsigma_j^\circ = r\lambda_j = \pm 2q^{1/2}$ *exactly*, a pure string-direction shift with no tail — proved twice, once from the sources' own formula and once by a spectral argument on the product that needs no transform at all. An $H^0$ shift changes no exponent, so $\nu_6(X \times \mathbf{P}^1, 0) = 4$ is a computed fact rather than a hypothesis, and the standing hypothesis is discharged at the endpoint rather than relocated. Arranging the weak factorization as a roof then makes every substitution a forward one, so no inverse is evaluated anywhere. What is left is a single statement about a single variety: that $X \times \mathbf{P}^1$ keeps at least one primitive-sixth eigenvalue along the accumulated displacement. Its $\nu_6$-carrying blocks are two copies of the cubic's rank-two Jordan block, which are exactly the hypotheses of Section 7, and the pencil along which they must be followed is polynomial per slot, so the engine to run there is Section 7 rather than a bulk gauge. Section 12 lists what the earlier versions got wrong; readers of those versions should start there.

# 1 The assertion at issue

References are to Section 4 of the draft, *Framed formal monodromy.* I name statements by their semantic labels as well as by number, because the edits proposed in Section 11 renumber the section: Definition 4.1 (`def:framed-sixth-multiplicity`), Definition 4.2 (`def:pro-laurent-gauge-group`), Lemma 4.3 (`lem:pv-base-change`), Lemma 4.5 (`lem:formal-base-shift`), Proposition 4.7 (`prop:framed-operations`), Lemma 4.9 (`lem:divisor-tagging`).

Lemma 4.3 ends with the sentence

> *If the gauge uses only integral powers of $z$ at every finite level, this conjugacy preserves the original-$z$-disc frame,*

whose proof is the single sentence "the deck action fixes every integral power of $z$ at each finite level, so the compatible gauge is single-valued for the framed turn." Lemma 4.5 then *defines*

$$M^{\mathrm{bulk}} = G\, M^{\mathrm{small,shifted}}_{\mathrm{f},V} G^{-1} \tag{4.1c}$$

and Proposition 4.7 concludes that "each center target summand has the framed monodromy of its specialized small connection." The conclusion does not follow from the display: conjugation preserves a characteristic polynomial for trivial reasons, but says nothing about whether the conjugated matrix is the intrinsic turn operator on horizontal solutions of the gauged module. What is missing is a transport lemma together with an algebra in which the turn acts on solutions.

## 2 The lemma over a differential field

Notation for $\mathrm{Exp}_V$, $K = \mathbf{C} \oplus V$ and $\Omega_V$ is the draft's, from (4.0) and (4.0a). Recall that $\mathrm{Exp}_V$ is defined on $(K, +)$, not on $\Omega_V$.

### The ground field

Let $\Gamma$ be a totally ordered abelian group and let

$$\mathcal{H} = \Omega_V(\!(\Gamma \oplus \mathbf{Z}e_z)\!)$$

be the Hahn field of series with well-ordered support and coefficients in $\Omega_V$, the value group ordered lexicographically with the $e_z$-coordinate *last*, as in the draft. Write $z = x^{e_z}$ and let $\theta$ multiply the coefficient at $(\gamma, k)$ by $k$; this is a derivation because $k$ is additive on the value group, and $\theta z = z$. Its constants are

$$\mathcal{C} := \mathcal{H}^{\theta=0} = \Omega_V(\!(\Gamma)\!).$$

Taking $\Gamma = 0$ gives $\mathcal{H} = \Omega_V(\!(z)\!)$. For $e \geq 1$ put $\mathcal{H}_e = \Omega_V(\!(\Gamma \oplus \frac{1}{e}\mathbf{Z}e_z)\!)$, a degree-$e$ extension containing $w = x^{e_z/e}$, and $\theta_w = e\theta$, so $\theta_w w = w$. All residues below are taken with respect to $\theta_w$ on the ramified side; the draft's $M_{\mathrm{RS},V}$ uses the same convention, and the two differ by the factor $e$ if one is careless.

### The solution algebra and the turn

**Definition 1.** Assume the exponential factors under consideration have *constant* coefficients, that is $\varphi \in w^{-1}\Omega_V[w^{-1}]$. The *universal formal solution algebra* $\mathcal{U}_e$ is the $\mathcal{H}_e$-algebra generated by symbols

$$w^\rho \ (\rho \in K), \qquad \ell, \qquad E(\varphi),$$

subject to $w^\rho w^{\rho'} = w^{\rho+\rho'}$, $w^n =$ the element $w^n$ of $\mathcal{H}_e$ for $n \in \mathbf{Z}$, $E(\varphi)E(\varphi') = E(\varphi + \varphi')$, $E(0) = 1$, with

$$\theta_w(w^\rho) = \rho\, w^\rho, \qquad \theta_w(\ell) = 1, \qquad \theta_w\big(E(\varphi)\big) = \theta_w(\varphi)E(\varphi).$$

The two hypotheses buried in that definition — constant coefficients in $\varphi$, and exponents $\rho$ in $K$ rather than in $\Omega_V$ — are not bookkeeping. The first is false for the modules the draft transports and is discussed at Remark 5; the second is forced because $\mathrm{Exp}_V$ is defined only on $K$, so a module whose residues leave $K$ needs Exp rebuilt over a larger field.

**Lemma 2** (Structure)**.** *Fix representatives $\Lambda \subset K$ for $K/\mathbf{Z}$ with $0 \in \Lambda$. Then $\mathcal{U}_e$ is free as an $\mathcal{H}_e$-module with basis $\{w^\rho \ell^m E(\varphi)\}$, $\rho \in \Lambda$, $m \geq 0$. In particular $\mathcal{U}_e \neq 0$ and $\mathcal{H} \subset \mathcal{U}_e$.*

*Proof.* One cannot argue that $K/\mathbf{Z}$ is torsion-free; it is not, since it contains $\mathbf{Q}/\mathbf{Z}$, and the rational classes are exactly the ones $\nu_6$ counts. Instead use base change. The group algebra $\mathcal{H}_e[K]$ is free over $\mathcal{H}_e[\mathbf{Z}] = \mathcal{H}_e[t, t^{-1}]$ on any set $\Lambda$ of coset representatives, and $\mathcal{U}_e$ is obtained from it by the base change $\mathcal{H}_e[t, t^{-1}] \to \mathcal{H}_e$, $t \mapsto w$, a surjection with kernel $(t - w)$ generated by a nonzerodivisor. Base change of a free module is free on the image of the same basis. The crossed multiplication $w^\rho w^{\rho'} = w^n w^{\rho''}$ is well defined, and the derivation is consistent with the integer identification because $w^n$ sits at $e_z$-exponent $n/e$ and $\theta_w$ multiplies it by $n$. The factors $E(\varphi)$ form the group algebra of a torsion-free group and $\ell$ is a polynomial variable. □

**Lemma 3** (The turn)**.** *There is a unique automorphism $\sigma$ of $\mathcal{U}_e$ with*

$$\sigma(w) = e^{2\pi i/e} w, \quad \sigma(w^\rho) = \mathrm{Exp}_V(\rho/e)\, w^\rho, \quad \sigma(\ell) = \ell + \tfrac{2\pi i}{e}, \quad \sigma\big(E(\varphi)\big) = E(\sigma\varphi),$$

*acting on $\mathcal{H}_e$ by multiplying the coefficient at $(\gamma, j/e)$ by $e^{2\pi i j/e}$. It fixes $\mathcal{H}$, hence $\mathcal{C}$, pointwise, and commutes with $\theta_w$.*

*Proof.* The action on $\mathcal{H}_e$ multiplies coefficients by a character of the value group, hence is a support-preserving ring automorphism, and is the identity on $\mathcal{H}$ because there $j \in e\mathbf{Z}$. Multiplicativity in $\rho$ is the homomorphism property of $\mathrm{Exp}_V$ on $(K, +)$, and the two prescriptions agree on integral exponents because $\mathrm{Exp}_V(n/e) = e^{2\pi i n/e}$ by (4.0a). Commutation with $\theta_w$ is not a diagonality argument, since neither $\sigma(\ell) = \ell + 2\pi i/e$ nor $\theta_w(\ell^m) = m\ell^{m-1}$ is diagonal; it is the direct check $\sigma\theta_w(\ell^m) = m(\ell + \frac{2\pi i}{e})^{m-1} = \theta_w\sigma(\ell^m)$, together with the diagonal cases. □

**Lemma 4** (Constants)**.** *Under the constant-coefficient hypothesis of Definition 1, $\mathcal{U}_e^{\theta_w = 0} = \mathcal{C}$.*

*Proof.* Let $v$ be the Hahn valuation of $\mathcal{H}_e$, additive because the coefficient field is a field; $\theta_w$ multiplies coefficients by integers and never enlarges a support, so $v(\theta_w c) \geq v(c)$.

Write $x = \sum c_{\rho,m,\varphi} w^\rho \ell^m E(\varphi)$ in the basis of Lemma 2 with $\theta_w x = 0$; the equation splits over $\varphi$. For $\varphi \neq 0$, let $M$ be the top $\ell$-degree in that block; for each $\rho$, $\theta_w(c_{\rho,M}) + \rho c_{\rho,M} + \theta_w(\varphi) c_{\rho,M} = 0$. Every monomial of $\theta_w(\varphi)$ has $\Gamma$-part 0 and strictly negative $e_z$-part, so $v(\theta_w(\varphi)c) < v(c) \leq v(\theta_w c + \rho c)$ for $c \neq 0$ and the lowest term cannot cancel; hence $c_{\rho,M} = 0$, and descending in $M$ kills the block.

For $\varphi = 0$ the coefficient equations are $\theta_w(c_{\rho,m}) + \rho c_{\rho,m} + (m+1)c_{\rho,m+1} = 0$. Descending in $m$: the top-degree coefficient satisfies the homogeneous equation $\theta_w(c) + \rho c = 0$, and on each monomial this reads $(j + \rho)c = 0$ with $j \in \mathbf{Z}$, so $c = 0$ unless $\rho \in \mathbf{Z}$, that is unless $\rho = 0$. For $\rho = 0$ the top coefficient is $\theta_w$-constant and the next equation gives $Mc_{0,M} = -\theta_w(c_{0,M-1})$, whose right side has zero component in $e_z$-degree 0; comparing that component gives $c_{0,M} = 0$ for $M \geq 1$. What survives is $\rho = 0$, $m = 0$, $\varphi = 0$ with $\theta_w$-constant coefficient, and the $\theta_w$-constants of $\mathcal{H}_e$ are those supported in $e_z$-degree zero, namely $\mathcal{C}$. □

*Remark* 5 (The constant-coefficient hypothesis is necessary, and fails in the application). Allow $\varphi$ a Novikov coefficient of positive weight, $\varphi = x^\gamma w^{-1}$ with $\gamma \in \Gamma$, $\gamma > 0$ — the shape of every exponential factor the draft actually meets, since the eigenvalues $\lambda$ of $c_1\star$ have positive Novikov weight. Then $v(\theta_w\varphi) > 0$ in the coefficient-dominant order and the domination argument reverses. Worse, the equation $\theta_w(c) = \theta_w(\varphi)c$ is solved by

$$c = \sum_{n \geq 0} \frac{(-1)^n}{n!}\, x^{n\gamma} w^{-n},$$

whose support $\{(n\gamma, -n/e)\}_{n\geq 0}$ is increasing, hence well ordered, so $c \in \mathcal{H}_e$ and $cE(\varphi)$ is a $\theta_w$-constant outside $\mathcal{C}$. Lemma 4 is therefore false in that regime, and with it Proposition 6. The structural reason is worth stating on its own: *in the draft's coefficient-dominant order,* $\exp(-\lambda/z)$ *is already a unit of the coefficient field*, so adjoining $E(\varphi)$ as a free symbol duplicates an existing element and inflates the constants. In that order the exponential factors are not symbols at all, and the entire irregularity of the formal solution sits in its divergent unit part, which is exactly where the rate comparison of Section 3 lives.

## Framed operators and transport

**Proposition 6** (The framed operator is the matrix of the turn)**.** *Let $D$ be a differential module over $\mathcal{H}$, free of rank $n$, with connection matrix $\Omega$, and let $Y \in \mathrm{GL}_n(\mathcal{U}_e)$ satisfy $\theta Y = -\Omega Y$. Then* $\mathrm{Sol}(D) = \{v : \theta v + \Omega v = 0\} = Y\mathcal{C}^n$*, free of rank $n$ and $\sigma$-stable; the matrix $M$ with $\sigma(Y) = YM$ lies in* $\mathrm{GL}_n(\mathcal{C})$ *and is well defined up to* $\mathrm{GL}_n(\mathcal{C})$*-conjugacy. For $D$ over* $\Omega_V(\!(z)\!)$ *in Levelt–Turrittin form, $M$ is the draft's framed operator: in a solution basis adapted to one deck orbit of length $d$ the matrix of $\sigma$ is block-cyclic with the draft's intertwiners $A_i$ as blocks, and the cyclic-block determinant gives* $\chi(X) = \det(X^d I - U)$ *with $U = A_{d-1} \cdots A_0$, which is (4.0b).*

*Proof.* From $\theta(Y^{-1}) = Y^{-1}\Omega$ one gets $\theta(Y^{-1}v) = Y^{-1}(\theta v + \Omega v)$, so horizontality of $v$ means $Y^{-1}v$ has $\theta_w$-constant entries, and Lemma 4 identifies those with $\mathcal{C}$. Since $\sigma$ commutes with $\theta$ and fixes $\Omega$, it preserves $\mathrm{Sol}(D)$; so $M = Y^{-1}\sigma(Y)$ has entries in $\mathcal{C}$ and is invertible, being a product of invertibles, with inverse $\sigma(Y^{-1})Y$. Two solution matrices differ on the right by an element of $\mathrm{GL}_n(\mathcal{C})$, which conjugates $M$. □

**Theorem 7** (Frame transport)**.** *Let $D, D'$ be as in Proposition 6, with solution matrices $Y, Y'$, and let $G \in \mathrm{GL}_n(\mathcal{H})$ satisfy $\Omega' = G\Omega G^{-1} - \theta(G)G^{-1}$. Then $GY$ is a solution matrix for $D'$ and $\sigma(GY) = (GY)M$. Consequently*

$$M_{\mathrm{f},V}(D') = C_0^{-1} M_{\mathrm{f},V}(D)\, C_0 \qquad \text{for some } C_0 \in \mathrm{GL}_n(\mathcal{C}),$$

*so the two framed operators have the same characteristic polynomial in $\mathcal{C}[X]$ and the same multiplicity at every root of unity.*

*Proof.* $\theta(GY) = \theta(G)Y - G\Omega Y = -\Omega'(GY)$. Since $\sigma$ fixes $\mathcal{H}$ pointwise, $\sigma(G) = G$, so $\sigma(GY) = G\sigma(Y) = GYM$. Writing $Y' = (GY)C_0$ with $C_0 \in \mathrm{GL}_n(\mathcal{C})$ gives $M' = C_0^{-1}MC_0$. □

*Remark* 8 (Which object is $M_{\mathrm{f},V}$). The framed operator is an automorphism of the horizontal module; its matrix depends on a choice of *solution* frame and is well defined up to conjugacy over the constants. The conjugate $GMG^{-1}$ appearing in the draft's (4.1c) is not the matrix of the turn in any solution frame — $Y'$ differs from $GY$ by a constant matrix, not by $G$ — but the matrix of the transported automorphism read in the *module* frame of $D'$. Both readings are legitimate, and they give the same characteristic polynomial, but the draft should say which it means, since conflating them is exactly the step at issue.

*Remark* 9 (The hypothesis on $G$ is the whole content). Since $\sigma$ fixes $\mathcal{H}$ pointwise, $\sigma(G) = G$ is automatic once $G$ has entries in $\mathcal{H}$; what matters is that $G$ lies in $\mathcal{H}$ and not in $\mathcal{U}_e$. No hypothesis-free statement exists: $Y$ itself is a gauge over $\mathcal{U}_e$ carrying $D$ to the trivial module, whose framed operator is the identity. The rank-one case shows what survives: if $g \in \mathcal{H}^\times$ and $\theta(g)/g = \delta$, then each monomial gives $(k - \delta)g_{(\gamma,k)} = 0$, so $\delta \in \mathbf{Z}$; a gauge over $\mathcal{H}$ shifts residues only by integers, which $\mathrm{Exp}_V$ kills.

### The scissors: these hypotheses have no common instance in the application

Theorem 7 is true, and in the application it is empty. Its two hypotheses pull in opposite directions.

- If $\Gamma = 0$, so $\mathcal{H} = \Omega_V(\!(z)\!)$, then Levelt–Turrittin supplies solution matrices, but the pro-Laurent comparison gauge has unbounded negative $z$-order and is not in $\mathrm{GL}_n(\mathcal{H})$.
- If $\Gamma \neq 0$ with the draft's coefficient-dominant order, the gauge lies in $\mathcal{H}$, but $\mathcal{H}$ is not a formal Laurent series field over its constants and Levelt–Turrittin is unavailable. Concretely, the $e_z$-component of the Hahn valuation is not a valuation on $\mathcal{H}$: for $f = x^{(0,0)} + x^{(1,-100)}$ and $g = -x^{(0,0)}$ its values on $f$ and $g$ are 0 while its value on $f+g$ is $-100$. So there is no Newton polygon, no slope filtration and no Turrittin algorithm. Nor can one descend to the bounded-$z$-order elements: $f = 1 - zx^{-\gamma}$ has inverse $-\sum_{k\geq 1} z^{-k}x^{k\gamma}$, so they do not form a field.

An abstract Picard–Vessiot ring for a module over $\mathcal{H}$ does exist, with constants $\mathcal{C}$ after passing to the divisible hull. So the missing ingredient is *not* a solution algebra. It is a canonical Levelt–Turrittin normal form over $\mathcal{H}$, singling out which element of the differential Galois group deserves to be called the turn. That is the precise shape of the remaining problem, and it also shows why a purely valuation-theoretic argument cannot supply it.

## 3 The receiver

The draft's receiver, in the proof of Proposition 4.7, is

$$\mathscr{R}_j = \Omega_{V,j} \otimes_{H_{0,j}} H_{\mathrm{tot},j}, \qquad H_{\mathrm{tot},j} = \mathbf{C}(\!(\Gamma_j^{\mathrm{coeff}} \oplus \mathbf{Z}e_z)\!). \tag{4.4c}$$

### Why the tensor product cannot simply be dissolved

The map $a \otimes h \mapsto a \cdot h$ into the Hahn field with $\Omega_{V,j}$ coefficients is not well defined: the copy of $H_{0,j}$ inside $\Omega_{V,j}$ consists of degree-zero constants, the copy inside $H_{\mathrm{tot},j}$ of monomial series, and the tensor product exists precisely to identify them. Nor can the two copies be assigned different jobs, with module coefficients read as constants and gauge bookkeeping as monomials: the gauge relation $\Omega' = G\Omega G^{-1} - \theta(G)G^{-1}$ ties $G$ and $\Omega$ through the same Novikov variables, so one copy is forced. With the constants reading the gauge leaves the field; with the monomial reading the exponential factors become $\Gamma$-supported and Remark 5 applies.

Two things should also be said about $\mathscr{R}_j$ itself. It is a tensor product of two field extensions and need not be a domain, and the draft reads characteristic polynomials and algebraic multiplicities in $\mathscr{R}_j[X]$; that deserves an explicit remark, which the draft partly gives by proving each factor injects.

### A better receiver

Extend the $\Gamma$-valuation of $H_{0,j}$ to its algebraic closure and then to $\Omega_{V,j}$, and take the field of series $\sum_k c_k z^k$ with $c_k \in \Omega_{V,j}$ whose support $\{(v_\Omega(c_k), k)\}$ is well ordered in the chosen order. This has a single copy of the coefficients, is a field rather than a tensor product of two fields, hosts the pro-Laurent gauge, and hosts the exponential factors as honest units. It is strictly better than (4.4c). It does not by itself supply Levelt–Turrittin, and it does not remove the rate criterion below.

## The two rates

Weight a monomial of coefficient weight $d$ and $z$-power $k$ by $a\,d + b\,k$, $a, b > 0$, and let $\varepsilon$ be the minimal weight of a generator of $J_j$.

- *Pro-Laurent gauges gain one filtration unit per unit of $z$-pole.* By the recursion (4.1a) and the draft's own conclusion, $G_\alpha$ has no power of $z$ below $-|\alpha|$ and a term of bulk degree $m$ lies in $F^m B$; every monomial of $J_j^N$ has weight at least $N$. Their weights grow like $m(a-b)$, which together with the draft's finite-below argument gives well ordered support when $a > b$.

- *Splitting gauges lose $w(\Delta\lambda)$ per unit of $z$-power.* This is the draft's own construction (4.9d): at order $n$ the block off-diagonal coefficient is removed by a Sylvester equation $[K_0, A_n] = B_n$ with $K_0$ the leading matrix, so one division by an eigenvalue difference per order, and $w(A_n) \geq -n\,w(\Delta\lambda)$. The draft's displayed cubic entry $-14/(81r^2)$ at $z$-order two exhibits the rate: one factor of $r^{-1}$ per unit of $z$-power. Three caveats. After a ramification $z = w^e$ the divisions occur once per $w$-order, so the loss is $e\,w(\Delta\lambda)$ per unit of $z$-power; at the cubic the draft proves the exponential blocks are unramified in $z$, so $e = 1$ there. The claim that the regular-singular normalization costs nothing needs the homogeneity argument that residue eigenvalues are rational, verified in the cubic block where $\det L_s = (s+1/6)(s+5/6)$. And $w(\Delta\lambda)$ is not controlled by $w(\lambda)$ when the Picard rank exceeds one, since $w$ and the quantum grading are different functionals and leading terms can cancel.

Both conditions hold together iff

$$\boxed{e\,w(\Delta\lambda) < \varepsilon}$$

## The criterion is not an artifact of choosing an order

For $C := e\,w(\Delta\lambda)/\varepsilon \geq 1$ the obstruction is not about well ordering at all. The $z^n$-coefficient of the single product $GP$ of a pro-Laurent gauge with a splitting gauge is $\sum_m g_m p_{n+m}$, whose terms have weight at least $m(1-C) - nC$; for $C \geq 1$ infinitely many terms have bounded weight, so that coefficient is an infinite sum in no completion, ordered or not. Any repair that forms this product must therefore establish $C < 1$. A repair that never forms it — see route 2 below — is not bound by this.

## The criterion depends on a normalization the draft may choose

This is the practical point, and it reverses the recommendation of the first version of this memo. The draft fixes $w(u) = 1$ and $w(s_{j,\ell}) = 1$ independently of $L$, and sets $w(Q^{i_* d} u^{\rho_C \cdot d}) = L(H \cdot i_* d) + \rho_C \cdot d$. Hence $\varepsilon = 1$ always, while the weights of the Novikov generators grow linearly in $L$. The criterion is therefore

$$e\big(L(H \cdot i_* d_0) + \rho_C \cdot d_0\big) < c_1 \cdot d_0,$$

using that the eigenvalues of $c_1\star$ are homogeneous of quantum degree one when $Q^d$ has degree $c_1 \cdot d$ — a convention I have checked against the draft's own low-dimensional vanishing proof, where the $Q^\beta$-component of $c_1\star$ maps $H^i$ to $H^{i+2-2c_1\cdot\beta}$, and against its explicit cubic block, where $K_0$ has eigenvalues $\pm 6r$, 0, 0 with $r = (3q)^{1/2}$ and $c_1 \cdot \ell = 2$. Consequences:

- Under the draft's instruction to choose $L$ "so large that" separation holds, the criterion fails at essentially every comparison with two distinct exponential factors — including the cubic endpoint.
- Under the smallest admissible $L$ it can hold. For a projective bundle with $c_1(V)\cdot\ell=0$ one may take $L=1$, giving $w(Q^\ell)=1=\varepsilon$ and $w(\Delta\lambda)=w(q^{1/2})=1/2$.
- Independently of $L$: separation forces $w\geq\varepsilon$ on every generator of $J_j$, and the image of $Q^{d_0}$ is a generator, so $c_1\cdot d_0\geq 2$ is necessary under any admissible weight. Index one fails outright.

So the cheapest available repair is to choose the separating weight minimally rather than large, and to check the resulting inequality case by case. The first version of this memo asserted the opposite, on the false ground that the ratio was scale-invariant; rescaling $w$ does move both rates equally, but $L$ is not a rescaling, because the bulk generators' weights are pinned at one.

# 4 The iterated receiver

The criterion above is a consequence of one architectural choice, and a reader of the second version of this memo proposed the alternative: the draft puts the intrinsic Novikov variables of the endpoint and the new completion variables into a single ordered group, which is what gives an eigenvalue difference positive weight. Nothing forces that. Treat the intrinsic Novikov fraction field as *coefficients*, and complete only in the new exceptional and bulk directions afterwards.

Two facts make the proposal concrete, and both are visible in the draft.

- *The unbounded negative $z$-order is bulk-driven, not Novikov-driven.* In the recursion (4.1a) the coefficient $G_\alpha$ has no power of $z$ below $-|\alpha|$, where $\alpha$ indexes the *bulk* parameter $\eta$, and the filtration depth is acquired by substituting $\eta\in F^1B$. A completion in the bulk directions alone therefore already hosts the pro-Laurent gauge; the Novikov directions are not needed for it.
- *With the Novikov directions as coefficients, the pathology disappears and the criterion is satisfied rather than evaded.* Eigenvalues of $c_1\star$ become constants of weight zero, so $\exp(-\lambda/z)$ is not a unit of the receiver but a genuine exponential symbol, the constant-coefficient hypothesis of Definition 1 holds, and Remark 5 evaporates. Since $w(\Delta\lambda)=0$, the inequality $e\,w(\Delta\lambda)<\varepsilon$ holds trivially, and the product $GP$ of Section 3 converges coefficientwise.

The shifts that do involve the exceptional Novikov root do not force those directions back into the filtration. The rank-one unit twist attached to $-(c-1)\lambda_j$, the fixed divisor shift, and the ambient target's $\tau^\circ=O(q^{-1})$ all enter Lemma 4.5 as its $a_0$ and $a_2^\circ$ terms, which are required only to be *constant in* the positive filtration, not to lie in it. With the Novikov directions as coefficients, $\exp(-a_0/z)$ is an honest exponential symbol rather than the object the draft must argue about separately.

### What the sources actually permit

I have checked this against the pinned sources rather than left it as a question, and the answer is positive for the comparison and negative for the gauge.

Iritani's Theorem 5.18 states $\Psi$ as an isomorphism of $\mathbf{C}[z]((q^{-1/s}))[[Q,\widetilde{\tau}]]$-modules, with the change of variables defined over $\mathbf{C}((q^{-1/(r-1)}))[[Q]]$. So the ring is *Laurent* in the exceptional root and *formal* in the ambient Novikov variables and the bulk. Two consequences.

*The comparison survives.* Inverting the Novikov monomials and completing in the exceptional and bulk directions is a ring homomorphism out of that ring. $\Psi$, its inverse, and the two inverse identities are module maps over it, so they base-change along it unchanged, and the framed operators' entries lie in the coefficient field, which injects. Nothing in the transport step needs Iritani's proof, only his statement, so the formality of his ring in $Q$ is not by itself an obstacle.

*The gauge does not.* The source's own reconstruction writes $\tau(\widetilde{\tau}) = \tau^\circ + t(\widetilde{\tau})$ and $\varsigma_j(\widetilde{\tau}) = \varsigma_j^\circ + s_j(\widetilde{\tau})$, with $t(\widetilde{\tau}) \in H^*(X)((q^{-1}))[[Q,\widetilde{\tau}]]$ and $s_j$ of the same shape over $H^*(Z)$. The initial terms are harmless: $\tau^\circ$ and $\varsigma_j^\circ$ are by definition the values at $Q = \widetilde{\tau} = 0$, hence carry no Novikov dependence at all, and the exceptional direction remains a completion direction as the draft already has it. But $t$ and $s_j$ vanish only at $Q = \widetilde{\tau} = 0$, so at $\widetilde{\tau} = 0$ they retain pure-Novikov terms. Those have to be removed by the normalized gauge of Lemma 4.5, and that construction converges by topological nilpotence of the bulk shift. Inverting the Novikov directions destroys exactly that.

So the iterated receiver is not free, and the decisive question is now small and graded rather than general:

> Is the pure-Novikov part of the coordinate shift — $t$ and $s_j$ restricted to $\widetilde{\tau} = 0$ — confined to $H^0 \oplus H^2$?

If it is, the string and divisor equations absorb it into the $a_0$ and $a_2^\circ$ terms of Lemma 4.5, which are required only to be constant in the positive filtration, and the iterated receiver goes through for the ambient summand with $w(\Delta\lambda) = 0$. If it is not, the Novikov directions must stay inside the filtration, eigenvalue differences reacquire positive weight, and the criterion of Section 3 returns.

The grading answers this, and the answer is negative in general. Iritani's Theorem 5.18(5) states that $\tau(\widetilde{\tau})$ and $\varsigma_j(\widetilde{\tau})$ are homogeneous of degree 2, and his conventions are $\deg Q^d = 2c_1(X)\cdot d$ and $\deg \tau_i = 2 - \deg \phi_i$. A pure-Novikov term $Q^d\phi$ therefore satisfies

$$\deg \phi = 2 - 2\,c_1(X)\cdot d,$$

so $c_1\cdot d = 1$ contributes in $H^0$, $c_1\cdot d = 0$ in $H^2$, and $c_1\cdot d = -1, -2, \ldots$ in $H^4, H^6, \ldots$; classes with $c_1\cdot d \geq 2$ cannot contribute at all. Hence the required confinement to $H^0 \oplus H^2$ follows exactly when every contributing effective class satisfies $c_1\cdot d \geq 0$, and it is not automatic: Iritani's theorem is stated for an arbitrary smooth projective $X$, and after several blowups in a weak factorization there is no reason for the intermediate ambient variety to have nef anticanonical class.

So the iterated receiver does not by itself remove arbitrary intermediate fourfolds from the problem. It removes them under a semipositivity hypothesis on the ambient, which the weak factorization does not supply. This is worth stating plainly, because it is the second proposed repair that fails on inspection, and it fails for a reason the draft would have inherited silently.

For the center the same analysis applies, with the additional and unchanged difficulty that its Novikov map is the noninjective one, which is why divisor tagging exists.

The scope consequence is the reason to try this first. If the ambient endpoint is handled this way, no inequality is needed for arbitrary intermediate fourfolds — the case that looked least tractable. What remains is the center summands, and there the draft's own low-dimensional classification is restrictive: for nef canonical class it produces an integral-$z$ gauge to a regular-singular connection, so there is no exponential separation at all and $w(\Delta\lambda) = 0$ again; the remaining minimal cases are $\mathbf{P}^1$, $\mathbf{P}^2$, ruled surfaces, and point blowups. Proving primitive-sixth vanishing directly for those, in the coordinates Iritani's specialization actually produces, is a smaller problem than a general transport theorem.

# 5 The framing route, checked against the sources and closed

The second version of this memo proposed the following route and called it the probable repair. I state it as it was proposed, then report the three checks, then state what survives. The checks were run against arXiv:2411.02266 as cached, arXiv:2307.03696v4, and arXiv:2307.13555v3; the full verdicts, locators and file hashes are in the companion report *The three framing-compatibility checks, run against Iritani's decomposition.*

## The route as proposed

Everything above treats the pro-Laurent bulk gauge of Lemma 4.5 as unavoidable. It may not be. Hinault, Yu, Zhang and Zhang work with $F$-bundles on the formal $u$-disc, where a framing is a regular flat connection and, in a framing-flat trivialization, the meromorphic connection takes the normal form

$$\nabla_{u\partial_u} = u\partial_u + u^{-1}K + (\mu D + H).$$

Their Theorem 4.34 characterizes when two connections of that shape are gauge-equivalent under

$$\Phi(u) \in \mathrm{GL}(H[[u]]), \qquad \Phi_{ij}(u) \in k[c_1, \dots, c_r][[u]],$$

with $\Phi$ then unique given its value at $u = 0$; the conditions are that the leading operators $K$ be conjugate by a parameter-polynomial $\phi$, that the two $\mu$ agree, and that the diagonal entries of $H$ agree after conjugation modulo the parameter ideal. Their applications are uniqueness of the decomposition for projective bundles and for blowups, complementing the existence theorems of Iritani–Koto and Iritani.

The shape of that gauge is exactly what this memo has been missing. A gauge in $\mathrm{GL}_n(K[[u]])$ has only nonnegative integral powers of the loop coordinate, adjoins no root of $u$, and is therefore fixed by the turn $u \mapsto e^{2\pi i}u$. Taking $\Gamma = 0$, so that $\mathcal{H} = \Omega_V((u))$ and Levelt–Turrittin is available, Theorem 7 applies with no further work:

$$\Phi \in \mathrm{GL}_n(K[[u]]), \quad \nabla' = \Phi\nabla\Phi^{-1} - \theta(\Phi)\Phi^{-1} \implies M_{\mathrm{f},V}(\nabla') \text{ and } M_{\mathrm{f},V}(\nabla) \text{ conjugate.}$$

No Hahn receiver, no product $GP$, and no inequality $e\,w(\Delta\lambda) < \varepsilon$, provided the decomposition of Proposition 4.7 can be identified with an $F$-bundle isomorphism of the class Theorem 4.34 governs. That identification, the matching of the $u = 0$ datum, and the separation of the center specialization were the three checks.

## Check one: the identification does not exist as posed

It fails three times over, and the first two are independent of the draft.

*Domain.* Section 4.4 of Hinault–Yu–Zhang–Zhang is titled *Equivalence of $F$-bundles over a point*, and both of its applications evaluate at the closed point $q = t = 0$ of the base, the large-radius limit; the proof of their Theorem 5.16 opens by observing that there the quantum product reduces to the classical cup product, so that the leading operator $K_{\mathrm{split}}$ of their (5.18) is a cup-product operator and hence nilpotent. The draft's $\nu_6$ is not evaluated there. It is read where the eigenvalues of $c_1\star$ separate — at the cubic, where the block carries $r = (3q)^{1/2}$ and the indicial roots $\pm 1/6$, $\pm 5/6$. At $q = 0$ that separation collapses and the exponential blocks the framed operator is built from do not exist. Their Theorems 5.20 and 5.24 do reconstruct the isomorphism over the whole base from its restriction at the limit point, but that is uniqueness of a map whose existence is Iritani's and Iritani–Koto's theorem; it is not a transport of framed monodromy between base points.

*Hypotheses, for the blowup.* Section 4.4 assumes the generalized eigenspaces of $K$ all have the same dimension, and the matrix calculus of Definition 4.33 and Theorem 4.34 rests on the resulting splitting $H_0^{\oplus m} \cong H$. The blowup decomposition is $H^*(X) \oplus \bigoplus_{i=1}^{m-1} H^*(Z)[-2i]$, whose summands differ in dimension whenever $\dim H^*(Z) \neq \dim H^*(X)$; the authors write a general endomorphism there as a matrix with $\Phi_{1,1} \in \mathrm{End}(H^*(X))$ and $\Phi_{i,i} \in \mathrm{End}(H^*(Z))$, so the mismatch is explicit in their own notation. Their blowup statement, Theorem 5.22, does assert existence of the isomorphism, and Theorem 5.24 gives the corresponding uniqueness; the point is not that it is unproved but that it is not *derived from* Theorem 4.34, whose hypotheses do not cover the unequal block sizes, and that its existence is referred to the earlier decomposition work. So Theorem 5.22 supplies no leverage of its own on the blowup half of Proposition 4.7 — equation (4.3), the half the headline theorem cannot do without — beyond what Iritani's theorem already gives. The projective-bundle half is genuinely covered: there the eigenspaces are $m$ copies of $H^*(X)$ by their Lemma 5.8, matching Section 4.4, and Theorem 5.16 is proved.

*Purpose.* Even where Theorem 4.34 applies, its conclusion compares the source at its base point with the target based at a shifted class $\Delta(a)$, pinned by their (5.17) for projective bundles and (5.23) for blowups. That shift is exactly what the pro-Laurent gauge of Lemma 4.5 exists to undo. The route relocates the bulk displacement rather than removing it.

The two normalizations do agree on where the displacement sits, which is a useful check on the draft's reading of the sources. In their limit-point coordinates the eigenvalue $\lambda_i \in \mathbf{C}[c_1, \ldots, c_m]$ is cohomology-valued, so $\Delta(a)$ acquires components in $H^{\geq 4}$ from the higher Chern classes, while the $H^2$ components cancel identically — their Lemma 5.8(2) gives $m\lambda_i \equiv m\xi^{i-1} - c_1 V$, cancelling the $c_1 V$ on the right of (5.17) — and the $H^2$ part of $\Delta(a)$ is left free. In Iritani's coordinates the same data reads $\varsigma_j^\circ = -(c-1)\lambda_j + h_{Z,j} + O(q^{-1/(c-1)})$ with $\lambda_j$ a *scalar* multiple of $q^{1/(c-1)}$, so the $H^0$ part is that scalar, the $H^2$ part is $h_{Z,j}$, and everything of degree at least four has moved into the tail. Same content, different bookkeeping; the draft's split of $\varsigma_j^\circ$ into unit twist, fixed divisor, and positive-filtration tail is right. The tail is irreducible: for a trivial rank-two bundle all $c_i(V)$ vanish and their $\Delta(a)$ is purely $H^0 \oplus H^2$, yet Iritani–Koto's $\varsigma_j^\circ$ still carries its $O(q^{-1/r})$ tail, so even $X \times \mathbf{P}^1$ does not escape the bulk gauge.

### Check two: the $u = 0$ datum, which passes

Their Theorems 5.20(2) and 5.24(2) determine the base map from its restriction to the base point up to a multiplicative constant in each logarithmic direction; the proof of their Proposition 4.31 shows why, since $df$ is recovered through $\Psi(d \log p_i) = d \log f_i$ and integrating a logarithmic form leaves one constant. That ambiguity is a character of the Novikov lattice: $q \mapsto cq$ is $Q^d \mapsto c^{E \cdot d} Q^d$, which is the divisor substitution (4.1) of the draft with $a_2^\circ$ a logarithmic multiple of the dual class. The draft already proves such a substitution is a $z$-constant coefficient automorphism and leaves $\nu_6$ unchanged, and already checks it is well defined on the image monoid. The bundle map carries no ambiguity at all once its restriction is fixed.

The ambiguity also never has to be tolerated, because the sources pin it. Iritani–Koto's (5.11) gives $\varsigma_j^\circ = r\lambda_j + [z^{-1}]\log\big(q^{c_1(V)/(rz)}F_j(1)\big)$ in closed form and their (5.12) gives $\Phi_j^\circ$; Iritani's Section 5.8.1 fixes $\Psi|_{Q=\widehat{\tau}=0}$ from his Lemma 5.13 and the explicit $\lambda_j$. One manuscript sentence citing those, and noting that any residual constant is a divisor substitution already covered by (4.1), settles this point for good, whichever repair eventually lands.

## Check three: the center specialization, which stays separate

It stays separate more firmly than the check anticipated. Their $A$-model $F$-bundle is built throughout on the one-variable Novikov projection $q^\beta \mapsto q^{\beta\cdot\omega}$ of their (2.21), and in the blowup application the center copies are the maximal $F$-bundles of $Z$ associated to $\sigma^*\omega_X$ — the Novikov direction pulled back from the ambient variety, which is the draft's specialization in a coarser form. No statement of theirs can therefore certify that a specialized center module carries an intrinsic framed monodromy. Divisor tagging remains the route, in the current manuscript, for identifying the specialized center contribution with the intrinsic invariant; route 3 of the list below bypasses that identification instead, by proving specialized vanishing directly in low dimension.

One genuine gain came out of it. Their Lemma 2.24 proves that the collapsed potential determines the full genus-zero potential, by expanding in the $H^2$ bulk directions and separating curve classes by their intersection numbers with an $H^2$ basis, under the finiteness of their Assumption 2.22. That is the mechanism of `lem:divisor-tagging`, in print, in a closely related setting. It does not close that lemma — it works at the level of the potential over a base retaining all $H^2$ directions, whereas the draft needs the separation at the level of a framed operator at a fixed bulk point, and the draft's collapse is by the pair $(i_*, \rho_C\cdot)$ rather than by a single nef class — but it makes the shape of the argument a cited one.

## What survives, and the corrected diagnosis

Two things survive, and both are worth more than the route was.

*The comparison isomorphisms need no receiver.* Iritani's $\Psi$ is an isomorphism of $\mathbf{C}[z]((q^{-1/s}))[[Q,\widehat{\tau}]]$-modules (Theorem 5.18), and Iritani–Koto's $\Phi$ is induced by a $\mathbf{C}[z,q][[Q,\widehat{\tau}]]$-module map and is an isomorphism over $\mathbf{C}[z]((q^{-1/r}))[[Q,\widehat{\tau}]]$ (proof of their Theorem 5.1(6)). Both are polynomial in $z$ with inverses in the same ring, hence lie in $\mathrm{GL}_n(F((z)))$ for $F$ the Laurent coefficient field, hence are fixed by the turn. Theorem 7 applies to them with $\Gamma = 0$, where Levelt–Turrittin is available: no Hahn field, no product $GP$, no inequality. The receiver problem was never a property of the comparison.

*The pro-Laurent gauge is not the draft's choice.* This corrects the sentence the second version of this memo built its recommendation on. Iritani–Koto transport along the same object: their reconstruction in Section 5.8 uses $M = \bigoplus_j e^{-\varsigma_j^\circ/z} M_B(\varsigma_j^\circ + s_j; Qq^{-c_1(V)/r})$, described in their footnote as a fundamental solution for the summands with respect to $(Q, s_0, \ldots, s_{r-1})$ but not for $q$ and $z$, satisfying $M = \mathrm{id} + O(z^{-1})$ and lying in $\mathrm{End}[z^{-1}]((q^{-1/r}))[[Q,s]]$. Per bulk degree it is a *polynomial* in $z^{-1}$ — which is the draft's own bound (4.1a), that $G_\alpha$ has no power of $z$ below $-|\alpha|$ — and it is unbounded only after summing over bulk degrees. The sources meet this object and handle it not by an ordered receiver but by Birkhoff factorization, splitting the $z$-regular factor $\Phi$ from the $\mathrm{id}+O(z^{-1})$ factor $M'$ in $(\Phi^\circ)^{-1}\circ M = M' \circ \Phi^{-1}$. Any repair should be measured against that, not against a receiver.

### The residual gap

After the two survivals, the blocker is no longer transport across the comparison. It is one statement:

> Let $T$ be smooth projective with quantum connection $\nabla$, and let $\tau^\bullet$ be a bulk parameter in the positive filtration — Iritani's $\tau^\circ = q^{-1}[Z] + O(q^{-2})$ on the ambient summand, the $O(q^{-1/(c-1)})$ tail of $\varsigma_j^\circ$ together with $s_j$ on a center summand. Show that the framed formal monodromy of $\nabla|_{\tau=\tau^\bullet}$ has the same primitive-sixth multiplicity as $\nabla|_{\tau=0}$.

Everything else in Proposition 4.7 is proved or reduced to a regular gauge, and `lem:divisor-tagging` inherits this gap and nothing else. Section 7 proves this statement whenever the block carrying $\nu_6$ is a rank-two Jordan block that is regular singular after one shearing, which covers the cubic ambient summand. What is still open is the same statement for an arbitrary block, which is what a center-by-center treatment would require.

# 6 The blockwise route, and what the flatness identities already give

This section carries out the second route far enough to see what it costs. It produces the exact evolution of the leading block operator under bulk motion and proves that multiplicity-one blocks carry no framed monodromy at all; for coalesced blocks it does not reduce the problem, but identifies why the full formal normal form, rather than the compressed pair $(U_i, \mu_i)$, is what has to be controlled. Conventions are Iritani–Koto's, from their Theorem 5.1: write $U = E\star_\tau$, $C_a = \phi_a \star_\tau$, and

$$\nabla_a = \partial_a + z^{-1} C_a, \qquad \nabla_{z\partial_z} = z\partial_z - z^{-1}U + \mu,$$

with $\mu$ the constant grading operator. Horizontal sections satisfy $z^2 \partial_z Y = (U - z\mu)Y$.

### The two identities

**Lemma 10** (Flatness)**.** *$[U, C_a] = 0$ and $\partial_a U = C_a + [C_a, \mu]$.*

*Proof.* The first is commutativity and associativity of the quantum product. For the second, expand $[\nabla_a, \nabla_{z\partial_z}] = 0$. The $z^{-2}$ term is $z^{-2}[U, C_a]$ and vanishes by the first identity; the $z^{-1}$ term is $z^{-1}\big(-\partial_a U + C_a + [C_a, \mu]\big)$, using $z\partial_z(z^{-1}C_a s) = -z^{-1}C_a s + z^{-1}C_a z\partial_z s$ and $\partial_a \mu = 0$. There is no $z^0$ term. $\square$

So bulk motion is not an arbitrary negative-loop gauge. The first identity says $C_a$ commutes with every spectral projector $P_i$ of $U$, hence preserves each generalized eigenspace $H_i$ with $U|_{H_i} = u_i + N_i$; the second says the motion of $U$ itself is expressed through $C_a$ and $\mu$ alone.

### The block frame

Let $T_a = \sum_j (\partial_a P_j) P_j$ be Kato's transport operator, so that $\partial_a P_i = [T_a, P_i]$ and $P_i T_a P_i = 0$. It is independent of $z$ and has no pole, so it moves the blocks without touching the loop coordinate. For an operator $A$ write $A_i = P_i A P_i$ and $A_{ij} = P_i A P_j$, and let $D_a A_i = \partial_a A_i - [T_a, A_i]$ be the covariant derivative in this frame. A short computation using $\partial_a P_i = [T_a, P_i]$ gives the general rule

$$D_a A_i = P_i(\partial_a A)P_i + P_i[A, T_a]P_i. \tag{6.1}$$

**Theorem 11** (Block evolution)**.** *Assume the eigenvalues $u_j$ of $U$ are pairwise distinct as scalars, so that each difference $u_i - u_j$ is invertible. Then, unconditionally,*

$$D_a U_i = C_{a,i} + [C_{a,i}, \mu_i], \qquad D_a \mu_i = \sum_{j\neq i} \big(\mu_{ij} X_{ji} - Y_{ij}\mu_{ji}\big),$$

*where $X_{ji} = P_j(\partial_a P_i)P_i$ and $Y_{ij} = P_i(\partial_a P_j)P_j$ are the unique solutions of the Sylvester equations*

$$\mathcal{S}_{ji}(X_{ji}) = -(\partial_a U)_{ji}, \qquad \mathcal{S}_{ij}(Y_{ij}) = -(\partial_a U)_{ij}, \qquad \mathcal{S}_{ji}(X) := (u_j - u_i)X + N_j X - X N_i,$$

*with $(\partial_a U)_{ji} = C_{a,j}\mu_{ji} - \mu_{ji}C_{a,i}$ and $(\partial_a U)_{ij} = C_{a,i}\mu_{ij} - \mu_{ij}C_{a,j}$. If* both *blocks are semisimple, $N_i = N_j = 0$, then $\mathcal{S}_{ji}$ is multiplication by $u_j - u_i$ and the formula collapses to*

$$D_a \mu_i = [\, C_{a,i}, S_i \,], \qquad S_i := \sum_{j\neq i} \frac{\mu_{ij}\mu_{ji}}{u_i - u_j}. \tag{6.2}$$

*Proof.* For $U$: by (6.1) and Lemma 10, $D_a U_i = P_i(C_a + [C_a, \mu])P_i + P_i[U, T_a]P_i$. Since $C_a$ commutes with $P_i$, the first term is $C_{a,i} + [C_{a,i}, \mu_i]$; since $U$ commutes with $P_i$, the second is $[U_i, P_i T_a P_i] = 0$. No perturbation formula is used, so this identity holds whatever the nilpotent parts.

For $\mu$: $\partial_a \mu = 0$, so (6.1) gives $D_a \mu_i = P_i[\mu, T_a]P_i$, which is the displayed sum. To evaluate $X_{ji}$, differentiate $[U, P_i] = 0$ to get $[\partial_a U, P_i] + [U, \partial_a P_i] = 0$ and compress by $P_j(\cdot)P_i$ with $j \neq i$: the first bracket contributes $(\partial_a U)_{ji}$ and the second contributes $U_j X_{ji} - X_{ji} U_i$, which is $\mathcal{S}_{ji}(X_{ji})$. The operator $\mathcal{S}_{ji}$ is invertible because $u_j - u_i$ is invertible and $X \mapsto N_j X - X N_i$ is nilpotent. The same argument with $i$ and $j$ exchanged gives $Y_{ij}$. When $N_i = N_j = 0$, substituting the two displayed expressions for $\partial_a U$ makes the terms containing $C_{a,j}$ cancel, leaving $\sum_{j\neq i}[C_{a,i}, \mu_{ij}\mu_{ji}]/(u_i - u_j)$. □

*Remark* 12 (The clean form does not cover the blocks that matter). An earlier version of this memo stated (6.2) under the weaker hypothesis $N_j = 0$ for the *neighbouring* blocks only. That was an error: differentiating $UP_i = (u_i + N_i)P_i$, the off-block part of $\partial_a(N_i P_i)$ contributes $X_{ji}N_i$, so the term $-XN_i$ cannot be dropped when the block $i$ itself is nonsemisimple. By Corollary 14 the only blocks that can carry $\nu_6$ are the coalesced ones, and in the draft's cubic calculation that block has $J_0 = \left(\begin{smallmatrix} 0 & 2 \\ 0 & 0 \end{smallmatrix}\right) \neq 0$. So $N_i \neq 0$ is the case of interest, and (6.2) does not apply to it. The cancellation that produced $S_i$ has to be redone with $\mathcal{S}_{ji}^{-1}$ in place of the scalar $1/(u_j - u_i)$; expanding $\mathcal{S}_{ji}^{-1} = \sum_{k\geq 0}(-1)^k (u_j - u_i)^{-k-1}(L_{N_j} - R_{N_i})^k$ shows the clean commutator is only the leading term of that expansion.

## Multiplicity-one blocks carry nothing

This is independent of the deformation and settles, in this normalization, the observation the earlier version of this memo only flagged.

**Theorem 13** (Simple blocks are invisible)**.** *Let $(\cdot,\cdot)$ be the Poincaré pairing. Then each $\mu_i$ is anti-self-adjoint for the restriction of the pairing to $H_i$, which is nondegenerate; consequently* $\operatorname{tr}\mu_i = 0$ *and the spectrum of $\mu_i$ is symmetric about* $0$*. If $u_i$ has multiplicity one, then $\mu_i = 0$, the block's formal solution is $e^{-u_i/z}$ times a single-valued formal unit, its framed monodromy is the identity, and it contributes no root of unity to $\nu_6$.*

*Proof.* Quantum multiplication is Frobenius, $(a \star b, c) = (a, b \star c)$, so $U$ is self-adjoint; distinct generalized eigenspaces of a self-adjoint operator are orthogonal, so each $P_i$ is self-adjoint and the pairing restricts nondegenerately to $H_i$. The pairing vanishes unless the degrees are complementary, so $\mu$ is anti-self-adjoint; hence so is $\mu_i = P_i \mu P_i$ on $H_i$. An anti-self-adjoint operator has $\operatorname{tr} \mu_i = \operatorname{tr} \mu_i^\dagger = -\operatorname{tr} \mu_i$ and characteristic polynomial even up to sign, giving the trace and symmetry statements. On a one-dimensional space anti-self-adjointness forces $\mu_i = 0$. Formal decoupling from the other eigenspaces by a gauge $g = 1 + z g_1 + \cdots$ can add terms $z\, c_1 + z^2 c_2 + \cdots$ to the scalar block, but it does not change its $z^0$ coefficient, since the only $z^0$ contribution is $-[g_1, U]_{ii} = [U_i, g_{1,ii}] = 0$ in rank one. So the decoupled block is $z^2 \partial_z y = (u_i + z^2 c_2 + \cdots) y$, whose solution is $e^{-u_i/z}$ times a formal unit with no $\log z$ and no fractional power: single-valued under the turn. $\square$

**Corollary 14.** *$\nu_6$ is carried entirely by the blocks of $E\star$ whose eigenvalue has multiplicity greater than one.*

This matches the draft's own cubic computation, where $K_0$ has eigenvalues $\pm 6r, 0, 0$ and the primitive-sixth exponents come from the repeated 0: the $\pm 6r$ blocks are simple, so by Theorem 13 they contribute the identity, and the draft's indicial roots $1/6$ and $5/6$ are $\pm 1/6$ modulo $\mathbf{Z}$, exactly the symmetry about 0 that the theorem forces on a block residue.

## What is left, exactly

**Proposition 15** (Isomonodromy criterion, semisimple blocks only)**.** *Suppose $N_i = 0$ and $N_j = 0$ for all $j$, so that the decoupled block connection is $\nabla_i = z\partial_z + A_i(z)$ with $A_i(z) = -z^{-1} U_i + \mu_i$ and* (6.2) *applies. Then $D_a \nabla_i$ is an infinitesimal gauge transformation by $\Theta_a(z) = -z^{-1} C_{a,i} + \beta$ if and only if*

$$[\beta, U_i] = 0 \qquad \text{and} \qquad [\beta, \mu_i] = [\, C_{a,i}, S_i \,].$$

*The $z^{-2}$ and $z^{-1}$ obstructions vanish identically for that choice of leading term; only the $z^0$ equation remains.*

*Proof.* Infinitesimally, $\delta A_i = [\Theta_a, A_i] - z\partial_z \Theta_a$. With $\Theta_a = z^{-1}\alpha + \beta$ this is $-z^{-2}[\alpha, U_i] + z^{-1}\big([\alpha, \mu_i] - [\beta, U_i] + \alpha\big) + [\beta, \mu_i]$. Under the stated hypotheses Theorem 11 gives $D_a \nabla_i = -z^{-1}\big(C_{a,i} + [C_{a,i}, \mu_i]\big) + [C_{a,i}, S_i]$. Taking $\alpha = -C_{a,i}$ kills the $z^{-2}$ term because $C_{a,i}$ commutes with $U_i$, and reduces the $z^{-1}$ equation to $[\beta, U_i] = 0$; the $z^0$ equation is the second displayed condition. $\square$

## Why this is not yet a reduction of the coalesced case

Proposition 15 is stated where it is true, and that is not where the problem is. Two things fail for a coalesced block, and the second is the deeper one.

First, by Remark 12, $S_i$ is not the right object once $N_i \neq 0$: the full Sylvester inverse has to be carried.

Second, and independently, the compressed pair $(U_i, \mu_i)$ does not determine the block's exponents. Formal decoupling of a nonsemisimple block from the other eigenspaces produces higher powers of $z$, and for a nilpotent leading term the Turrittin reduction reaches down into them. The draft's own cubic calculation is the demonstration. Its reduced rank-two block is

$$z^2 \partial_z \begin{pmatrix} \widetilde{S}_3 \\ \widetilde{S}_4 \end{pmatrix} = \big[ J_0 + z D_0 + z^2 E_0 + O(z^3) \big] \begin{pmatrix} \widetilde{S}_3 \\ \widetilde{S}_4 \end{pmatrix}, \qquad J_0 = \begin{pmatrix} 0 & 2 \\ 0 & 0 \end{pmatrix}, \quad D_0 = \begin{pmatrix} -19/18 & 0 \\ 0 & 19/18 \end{pmatrix},$$

with $E_0$ carrying the entry $-8/81$. The ansatz $\widetilde{S}_3 = z^\rho(1+O(z))$, $\widetilde{S}_4 = cz^{\rho+1}(1+O(z))$ gives $\rho = 2c - 19/18$ and $c(\rho+1) = \frac{19}{18}c - \frac{8}{81}$, hence $\rho^2+\rho+\frac{5}{36} = 0$ and $\rho = -\frac{1}{6}, -\frac{5}{6}$. The $z^2$ coefficient is not decoration: setting its $-8/81$ entry to zero leaves $c(\rho+1) = \frac{19}{18}c$, whose solutions are $\rho = 1/18$ with $c = 5/9$, and $\rho = -19/18$ with $c = 0$ — the classes $\pm 1/18$ modulo $\mathbf{Z}$, which are eighteenth roots of unity and contribute nothing to $\nu_6$. So a deformation theory that tracks only $U_i$ and $P_i \mu P_i$ cannot see the primitive-sixth exponents at all, and any proposed evolution theorem must reproduce this example. It is the regression test.

### The corrected program

1. Flatness preserves the spectral decomposition of $U$ and gives the exact evolution $D_a U_i = C_{a,i} + [C_{a,i}, \mu_i]$, unconditionally.

2. Multiplicity-one blocks contribute no cyclotomic monodromy (Theorem 13), so only coalesced blocks are in play.

3. For a coalesced block the off-block projector equations are governed by the full Sylvester operator $(u_j - u_i) + L_{N_j} - R_{N_i}$, invertible but not scalar.

4. What must be deformed is the entire formal block normal form $J + zD + z^2E + \cdots$, not the compressed pair, and the deformation theory has to reproduce the cubic $2\times 2$ example above.

A caution on the literature before it is leaned on. The extension results for isomonodromic deformations with coalescing eigenvalues assume, among other hypotheses, that the leading matrix is diagonalizable at the coalescence point. The carrier here is nonsemisimple at leading order — $J_0 \neq 0$ — so those theorems do not apply off the shelf, and the coalescence question stated below is not answered by citing them.

The risk that remains is unchanged in kind and now better located: a coalesced block can *split* under the displacement, the splitting being governed by $D_a U_i$, and if the rank-two block carrying $\pm 1/6$ split into simple blocks those would carry residue 0 by Theorem 13 and $\nu_6$ would drop by two. Proving that the reconstruction displacement does not do that, or does it in a count-preserving way, is the residual problem. Section 7 now does exactly that, by shearing the block before deforming it.

## 7 The coalesced block is rigid

This section closes the residual gap of Section 5 for a block of the cubic's type. The route is the one Section 6 said was needed: deform the *full* formal normal form rather than the compressed pair. The device that makes it finite is to shear first. After the shearing the block is regular singular, its residue already contains the $z^2$ coefficient that the regression test showed to be load-bearing, and — this is the point — the bulk connection loses its pole, so the residue moves by conjugation alone.

### Setting

Let $\Lambda$ be the coefficient field of the small connection, containing $r = (3q)^{1/2}$ and its inverse, and let $B = \Lambda[[\tau]]$ be the formal even bulk germ in finitely many even coordinates $\tau_a$. Write the quantum connection in the conventions of Section 6,

$$z\partial_z Y = \big(z^{-1}U - \mu\big)Y, \qquad \partial_a Y = -z^{-1}C_a Y,$$

with $U = E\star_\tau$, $C_a = \phi_a\star_\tau$ and $\mu$ constant, so that Lemma 10 holds: $[U, C_a] = 0$ and $\partial_a U = C_a + [C_a, \mu]$. Everything below is formal in $\tau$ and needs no completion of the loop coordinate, no ordered receiver, and no bulk gauge.

There is one standing hypothesis, verified for the cubic at $\tau = 0$:

(H1) $U(0)$ has an eigenvalue $u_0$ whose generalized eigenspace $H_0$ has rank two and on which the nilpotent part is nonzero; every other eigenvalue differs from $u_0$ by a unit of $\Lambda$.

For the cubic, (H1) holds with $u_0 = 0$ and the other eigenvalues $\pm 6r$.

Earlier versions of this memo carried a second standing hypothesis (H2): that after the shearing of Step 4 the block is regular singular at $\tau = 0$, that is $f(0) = 0$ in the notation there. It was verified for the cubic by inspection, because the draft's $D_0$ is diagonal. It is now a theorem for every rank-two coalesced block of every smooth projective target (Theorem 20), proved from the Frobenius property of quantum multiplication, and it holds pointwise on the germ rather than only at the base point. Nothing below assumes it.

### Step 1: decoupling, and the bulk connection stays block diagonal

**Lemma 16.** *There is a unique gauge $g = I + O(z)$ over $B$, block-diagonalizing the $z$-connection with respect to the generalized eigenspaces of $U$. In the gauged frame the bulk connection is block diagonal as well, and its leading coefficient on $H_0$ is $C_{a,0} = P_0 C_a P_0$.*

*Proof.* Existence and uniqueness of $g$ is the splitting lemma at Poincaré rank one, solved order by order by Sylvester equations whose operators are invertible because the eigenvalue differences are units. For the second statement write the gauged pair as $\widetilde{\mathcal{A}} = z^{-1}U + O(1)$, block diagonal, and $\widetilde{\mathcal{C}}_a = z^{-1}\check{C}_a(z)$ with $\check{C}_a = \check{C}^{(0)} + z\check{C}^{(1)} + \cdots$, and expand the flatness identity

$$\partial_a \widetilde{\mathcal{A}} = -z\partial_z \widetilde{\mathcal{C}}_a + [\widetilde{\mathcal{A}}, \widetilde{\mathcal{C}}_a]. \tag{8.1}$$

Its off-block part in bidegree $(i, j)$, $i \neq j$, has zero left side because $\widetilde{\mathcal{A}}$ is block diagonal. At order $z^{-2}$ this reads $(u_i - u_j)(\check{C}^{(0)})_{ij} = 0$, and at each successive order the same invertible factor multiplies the next off-block coefficient. So every off-block coefficient vanishes. Since $g = I + O(z)$, the leading term is unchanged. □

### Step 2: the exponential factor, and the commutant

Put $u_0(\tau) = \frac{1}{2}\operatorname{tr}\big(U|_{H_0}\big)$ and $N := U|_{H_0} - u_0 I$, so $N$ is traceless and $N(0) \neq 0$ is nilpotent. Twisting the block by $\exp(u_0/z)$ removes its exponential factor; this changes neither the regular-singular part nor the framed monodromy. The twist replaces the leading bulk coefficient $C_{a,0}$ by $C_{a,0} - (\partial_a u_0)I$.

**Lemma 17.** *$C_{a,0} = p_a I + q_a N$ with $p_a, q_a \in B$, and $p_a = \partial_a u_0$. After the twist the leading bulk coefficient on the block is $q_a N$.*

*Proof.* $[U, C_a] = 0$ gives $[U|_{H_0}, C_{a,0}] = 0$. A $2 \times 2$ matrix over a local ring whose reduction is non-scalar is regular, and $U|_{H_0}$ reduces to $u_0(0)I + N(0)$ with $N(0) \neq 0$; the commutant of a regular $2 \times 2$ matrix is free of rank two on $I$ and the matrix itself, hence equals $B \cdot I \oplus B \cdot N$. Taking traces in $\partial_a U = C_a + [C_a, \mu]$ on the block gives $2\partial_a u_0 = \operatorname{tr} C_{a,0} = 2p_a$. □

## Step 3: the eigenvalue never splits

**Theorem 18** (No splitting)**.** $\det N \equiv 0$ *on* $B$*. Consequently* $N$ *is nilpotent at every point of the formal germ, the block keeps a single eigenvalue of multiplicity two, and by (H1) it is a single Jordan block.*

*Proof.* Write $d := \det N \in B$; by (H1), $d(0) = 0$. From Theorem 11 the block leading operator evolves by $D_a U_0 = C_{a,0} + [C_{a,0}, \mu_0]$, so by Lemma 17

$$D_a N = D_a U_0 - (\partial_a u_0) I = q_a N + q_a [N, \mu_0].$$

For a traceless $2 \times 2$ matrix $\operatorname{adj}(N) = -N$ and $N^2 = -dI$, so

$$\partial_a d = \operatorname{tr}\bigl(\operatorname{adj}(N)\, D_a N\bigr) = -q_a \operatorname{tr}(N^2) - q_a \operatorname{tr}\bigl(N[N, \mu_0]\bigr) = 2 q_a d,$$

using $\operatorname{tr}(N[N, \mu_0]) = \operatorname{tr}(N^2 \mu_0) - \operatorname{tr}(N \mu_0 N) = 0$. Now suppose $d \neq 0$ and let $m \geq 1$ be the order of its lowest nonvanishing homogeneous part $d_m$ in $\tau$. The right side of $\partial_a d = 2 q_a d$ has order at least $m$, while the left side has order $m - 1$ with homogeneous part $\partial_a d_m$. Hence $\partial_a d_m = 0$ for every $a$, so $d_m = 0$, a contradiction. Note $d$ is basis independent, so no choice of block frame enters. □

This is the falsifier of Section 6 answered: the reconstruction displacement cannot split the block that carries $\nu_6$. It also shows the Kato frame is not needed for this step, since $\det N$ is frame independent.

## Step 4: shear

By Theorem 18, $N^2 = 0$ and $N \neq 0$, so $\ker N = \operatorname{im} N$ is a free rank-one submodule; choose a frame $(e_1, e_2)$ of $H_0$ with $N e_1 = 0$, $N e_2 = \nu e_1$, $\nu \in B^\times$, depending formally on $\tau$, and let $\Gamma_a$ be the resulting frame connection, which has no pole in $z$. Apply $S = \operatorname{diag}(1, z)$ and write the sheared pair as

$$z\partial_z \widehat{Y} = \widehat{\mathcal{A}} \widehat{Y}, \quad \widehat{\mathcal{A}} = z^{-1} A_{-1} + A_0 + z A_1 + \cdots, \qquad \partial_a \widehat{Y} = -\widehat{\mathcal{C}}_a \widehat{Y}, \quad \widehat{\mathcal{C}}_a = z^{-1} K_a + G_a + \cdots.$$

Conjugation by $S$ sends $E_{12} \mapsto z E_{12}$, $E_{21} \mapsto z^{-1} E_{21}$ and fixes diagonal matrices, and $-S^{-1} z \partial_z S = -\operatorname{diag}(0, 1)$. Writing the decoupled block connection before shearing as $z^{-1} N + A'_0 + z A'_1 + \cdots$, one gets

$$A_{-1} = f\, E_{21}, \quad f := (A'_0)_{21}, \qquad A_0 = R := \nu E_{12} + \operatorname{diag}\bigl((A'_0)_{11}, (A'_0)_{22} - 1\bigr) + (A'_1)_{21} E_{21}. \tag{8.2}$$

On the bulk side, the leading coefficient $q_a N = q_a \nu E_{12}$ shears *up* to order $z^0$, and the only surviving pole comes from the $(2, 1)$ entry of the regular part:

$$K_a = k_a E_{21}, \qquad k_a := \bigl(\check{C}^{(1)}_{a,0} + \Gamma_a\bigr)_{21}. \tag{8.3}$$

There is no $z^{-2}$ term, and this is where Lemma 17 does its work: a $(2, 1)$ entry in $C_{a,0}$ would have produced one, and the commutant forbids it.

For the cubic, (8.2) reproduces the draft's own numbers. With $\nu = 2$, $A'_0 = D_0 = \operatorname{diag}(-19/18, 19/18)$ and $(A'_1)_{21} = -8/81$,

$$f = 0, \qquad R = \begin{pmatrix} -19/18 & 2 \\ -8/81 & 1/18 \end{pmatrix}, \qquad \operatorname{tr} R = -1, \quad \det R = \tfrac{5}{36},$$

so the characteristic polynomial of $R$ is exactly $\rho^2 + \rho + 5/36$ and its roots are $-1/6$ and $-5/6$. The $z^2$ coefficient enters through $(A'_1)_{21}$, so the regression test of Section 6 is met by construction rather than by accident: the object being deformed below is $R$, which sees it.

### Step 5: the Poincaré pairing survives the decoupling, and kills $f$

The shearing of Step 4 produces a pole exactly when $f = (A'_0)_{21} \neq 0$. Earlier versions assumed $f(0) = 0$ as a standing hypothesis, verified for the cubic by inspecting the draft's diagonal $D_0$. It is not a hypothesis: it is forced by the Frobenius property of quantum multiplication, at every point of the germ and for every smooth projective target. The only thing needed is that the decoupling gauge of Step 1 preserves the pairing, which is what the following lemma supplies — the step the earlier list of open questions identified as missing.

Let $G$ be the Poincaré pairing on the even part, symmetric and nondegenerate. Quantum multiplication is Frobenius, so $U = E\star$ is $G$-self-adjoint, and $\mu$ is $G$-anti-self-adjoint. Writing $\mathcal{A}(z) = z^{-1}U - \mu$, those two facts are exactly

$$\mathcal{A}(-z)^T G + G\,\mathcal{A}(z) = 0. \tag{8.4}$$

**Lemma 19** (The decoupling gauge is an isometry)**.** *The unique gauge $g = I + O(z)$ of Lemma 16 satisfies $g(-z)^T G\, g(z) = G$. Hence the decoupled connection satisfies* (8.4) *with the same constant pairing, and so does each block with the restriction of $G$ to it, before and after the twist by* $\exp(u_0/z)$*.*

*Proof.* Let $\sigma$ be the involution $\mathcal{A} \mapsto -G^{-1}\mathcal{A}(-z)^T G$ on connection matrices, so that (8.4) reads $\mathcal{A}^\sigma = \mathcal{A}$. A direct computation with the gauge action $g \cdot \mathcal{A} = g^{-1}\mathcal{A}g - g^{-1}z\partial_z g$ gives

$$(g \cdot \mathcal{A})^\sigma = h \cdot (\mathcal{A}^\sigma), \qquad h := G^{-1}g(-z)^{-T}G,$$

the two terms matching separately: the conjugation term because $h^{-1} = G^{-1}g(-z)^T G$, and the logarithmic-derivative term because $h^{-1}z\partial_z h = -G^{-1}\big(z\partial_z g(-z)\big)^T g(-z)^{-T}G$. Take $g$ to be the decoupling gauge and put $\widetilde{\mathcal{A}} = g \cdot \mathcal{A}$. The generalized eigenspaces of a $G$-self-adjoint operator are $G$-orthogonal, so $G$ is block diagonal and $\widetilde{\mathcal{A}}^\sigma$ is block diagonal whenever $\widetilde{\mathcal{A}}$ is; and $h = I + O(z)$ because $g = I + O(z)$. So $h$ is a second gauge of the normalized form carrying $\mathcal{A}$ to block-diagonal form, and the uniqueness in Lemma 16 gives $h = g$, which is the claim. Restriction to a block is legitimate because $G$ restricts nondegenerately to each block. The twist by $\exp(u_0/z)$ subtracts the scalar $z^{-1}u_0 I$ from the block connection, and (8.4) is insensitive to that because the subtracted term is odd in $z$ and self-adjoint. $\square$

Expanding (8.4) for the twisted decoupled block connection $z^{-1}N + A'_0 + zA'_1 + \cdots$ coefficient by coefficient gives the parity rule

$$N^T G_0 = G_0 N, \qquad (A'_p)^T G_0 = (-1)^{p+1} G_0 A'_p, \tag{8.5}$$

that is, $N$ and the odd coefficients are $G_0$-self-adjoint and the even coefficients are $G_0$-anti-self-adjoint. The $p = 0$ case is the tracelessness statement of Section 6, now with the decoupling correction included rather than only for $P_i \mu P_i$.

**Theorem 20** (The sheared block is regular singular at every point)**.** *Under (H1), $f = (A'_0)_{21} = 0$ identically on $B$.*

*Proof.* By (8.5), $N$ is $G_0$-self-adjoint, and $N^2 = 0$ by Theorem 18, so for all $x, y$ one has $(Nx, Ny) = (x, N^2 y) = 0$: the image of $N$ is isotropic. In particular $(e_1, e_1) = 0$, and nondegeneracy of $G_0$ then forces $(e_1, e_2) \neq 0$. Again by (8.5), $A'_0$ is $G_0$-anti-self-adjoint, and for a symmetric form that gives $(A'_0 x, x) = 0$ for every $x$. Taking $x = e_1$ and expanding $A'_0 e_1 = \alpha e_1 + f e_2$ gives $f\,(e_2, e_1) = 0$, hence $f = 0$. $\square$

This is stronger than the deleted hypothesis in two ways. It holds pointwise on the germ, with no differential equation and no lowest-order argument, which is what a statement outside the germ needs; and it holds for every rank-two coalesced block of every smooth projective target, so the cubic's diagonal $D_0$ was not an accident of that example.

### Step 6: no irregularity appears, and the bulk connection is regular

**Theorem 21** (The sheared block stays regular singular)**.** *$k_a \equiv 0$ on $B$. Hence at every point of the formal germ the block is regular singular with residue $R$, and in the sheared frame the bulk connection has no pole in $z$.*

*Proof.* Expand (8.1) for the sheared pair. The coefficient of $z^{-2}$ is $[A_{-1}, K_a] = [fE_{21}, k_a E_{21}] = 0$, which is vacuous. The coefficient of $z^{-1}$ is

$$\partial_a(f)E_{21} = k_a E_{21} + f\,[E_{21}, G_a] + k_a\,\mathrm{ad}_R(E_{21}).$$

Writing $R = \begin{pmatrix} \alpha & \nu \\ \gamma & \delta \end{pmatrix}$ one computes $\mathrm{ad}_R(E_{21}) = \begin{pmatrix} \nu & 0 \\ \delta - \alpha & -\nu \end{pmatrix}$, whose diagonal is $\mathrm{diag}(\nu, -\nu)$ and does not involve $\gamma$. The left side is off-diagonal, so comparing diagonal entries gives $k_a\nu = -f\,\big([E_{21}, G_a]\big)_{11}$, and since $\nu$ is a unit, $k_a = f\,h_a$ for some $h_a \in B$. Comparing $(2,1)$ entries then gives

$$\partial_a f = f\Big(h_a\big(1 + \delta - \alpha\big) + \big([E_{21}, G_a]\big)_{21}\Big),$$

a linear equation $\partial_a f = f\,w_a$, which is now redundant: Theorem 20 already gives $f \equiv 0$ outright, and $k_a = f\,h_a$ then gives $k_a \equiv 0$. □

*Remark* 22 (What the flatness route still buys). The displayed equation $\partial_a f = f\,w_a$ is kept because it is the only place where the argument would survive the loss of the Frobenius property: with $f(0) = 0$ assumed rather than proved, the lowest-order argument of Theorem 18 applies verbatim and yields the same conclusion. In any setting where the Frobenius input is unavailable, or for a block of a connection that is not the quantum connection of a smooth projective target, that is the route to use, at the cost of carrying the hypothesis.

### Step 7: the exponents are constant

**Theorem 23** (Rigidity of the formal type)**.** *$\partial_a R = [R, G_a]$ for every bulk direction $a$. Hence the characteristic polynomial of $R$ has coefficients in $\Lambda$, constant in $\tau$; the formal monodromy of the block is conjugate to $\exp(2\pi i R)$ at every point of the formal germ; and the primitive-sixth multiplicity of the block is constant.*

*Proof.* By Theorem 21 the sheared pair has $A_{-1} = 0$ and $K_a = 0$. The coefficient of $z^0$ in (8.1) is $\partial_a A_0 = [A_{-1}, H_a] + [A_0, G_a] + [A_1, K_a]$, which collapses to $\partial_a R = [R, G_a]$. A matrix moving by an infinitesimal conjugation has constant characteristic polynomial: $\partial_a \det(XI - R) = 0$ entrywise. Since $\widehat{\mathcal{A}} = R + O(z)$, the block is regular singular and its formal monodromy is conjugate to $\exp(2\pi i R)$; shearing changed the exponents by integers only, which $\nu_6$ does not see. □

**Corollary 24** (The residual gap, closed at the cubic)**.** *For a smooth cubic threefold, the framed formal monodromy of the zero block has $\nu_6 = 2$ at every point of the formal even bulk germ, in particular at any bulk parameter $\tau^\bullet$ in the positive filtration. The statement of Section 5 therefore holds for the cubic ambient summand.*

*Proof.* Combine Theorem 23 with the computation after (8.2): the characteristic polynomial is $\rho^2 + \rho + 5/36$ everywhere, with roots $-1/6, -5/6$, whose monodromy eigenvalues $e^{\mp\pi i/3}$ are primitive sixth roots of unity. Since the coefficients of the characteristic polynomial are constants of $\Lambda$, evaluation at a topologically nilpotent $\tau^\bullet$ is legitimate and returns the same polynomial; no convergence of a bulk gauge is involved. □

### Three remarks

*Remark* 25 (Why this is not the compressed-pair argument). The deformed object is $R$ of (8.2), not the pair $(U_0, \mu_0)$. The shearing has folded the $z^2$ coefficient of the unsheared block into the residue, so the failure mode of Section 6 — deleting $-8/81$ and moving the exponents to $\pm 1/18$ — cannot occur here: that deletion changes $R$ itself, hence its characteristic polynomial, and Theorem 23 freezes exactly that polynomial.

*Remark* 26 (Relation to Cai's Proposition 6). Cai states the $\pm 1/6$ exponents for the *big* quantum connection and proves it by constructing $M = I + \sum M_n t^n$ with $\partial_{t_i} M = -z^{-1} P_i M$, so that $M^{-1}\nabla_{t_i} M = \partial_{t_i}$ and $M^{-1}\nabla_z M$ is bulk independent. That gauge is precisely the pro-Laurent object of Lemma 4.5 and of Iritani–Koto's Section 5.8: per bulk degree it is polynomial in $z^{-1}$, unbounded only after summing bulk degrees. Over $\Lambda((z))[[t]]$, with the bulk kept formal, that argument is sound and proves the same conclusion, because $M$ is fixed by the turn and Theorem 7 applies with the bulk variables as formal parameters. What it does not do by itself is license evaluation at a *specialized* bulk parameter with Novikov coefficients, which is where the receiver discussion of Section 3 came from. The argument above avoids the gauge altogether and produces a $\tau$-constant characteristic polynomial, so specialization is a substitution into constants.

### A shorter route to the endpoint, in the sources' own terms

Two paragraphs of Katzarkov–Kontsevich–Pantev–Yu's Example 6.21 bear directly on this and should be on the record, because between them they suggest a route to the one-stabilization theorem that needs neither this memo's transport lemma nor its rigidity theorem.

First, they compute the atomic composition of a smooth cubic threefold at the hyperplane point: two one-dimensional atoms for the nonzero eigenvalues of $\mathrm{Eu}\star$, and one atom $\alpha(X)$ for the eigenvalue 0. That is the block structure used throughout this memo, in their language. They add that "the Witt algebra argument from Remark 3.14 shows that there is no further splitting in a neighborhood of $b$" — which is Theorem 18, asserted rather than proved, and their Remark 3.14 introduces its finer clauses with "more fine results can be proved". Section 7 supplies proofs of both the splitting statement and the finer one, for this block type, from flatness alone.

Second, and more consequentially, they separate $\alpha(X)$ from a curve atom not by formal monodromy but by a Serre automorphism. They observe that $\alpha(X)$ looks like the atom of a smooth projective genus-five curve, then note that as a Hodge atom enhanced with its Serre automorphism $S$, the graded minimal polynomial forbids $S$ from having eigenvalue $-1$, so it cannot be the Serre automorphism of a genus-five curve. They conclude that every smooth cubic threefold is irrational, and remark that the same approach applies to other Fano threefolds.

Their $S^5 = [3]$ is quoted correctly above — it is what Example 6.21 says — but it is not the relation to use when computing on a K-group. Kuznetsov's fractional Calabi–Yau relation for the same category is $S^3 = [5]$, and on the numerical Grothendieck group, where a shift $[k]$ acts by $(-1)^k$, the two predict $S^3 = -I$ and $S^5 = -I$ respectively; the Serre operator computed from Riemann–Roch satisfies the first exactly and fails the second, its characteristic polynomial being $\Phi_6$ and its order six, in agreement with the exponents $-1/6, -5/6$ recorded after (8.2). The offset is systematic rather than a slip: their Example 6.20 gives $S^3 = [4]$ for the cubic fourfold, whose Kuznetsov component is a K3 category with $S = [2]$ categorically. The two are different objects — the Serre automorphism of a Hodge atom in the cohomological $\mathbf{Z}$-grading, and the categorical Serre functor on a K-group — so a citation should follow the object it is about. Locators and the full comparison are in '2026-08-15-c912-kkpy-imports-source-check.md'.

A second normalization difference runs alongside it and explains a substitution this lane had observed but not accounted for. Their statements are made after the half-parity gauge $u^g$ of Claim 6.15, this memo's count before it. That gauge shifts exponents by half the cohomological degree, multiplying monodromy eigenvalues by a parity-dependent sign: primitive sixth roots become primitive cube roots and $\pm 1$ swap. It is the $\lambda \mapsto -\lambda$ substitution between the prime-Fano classification's polynomial $R$ and the Serre side. Every separation used below distinguishes roots of unity of order at most two from roots of order three or six, which the gauge preserves.

The consequence for the endpoint is worth stating plainly. Their argument is dimension three, where Proposition 5.30 requires excluding only points and curves. The one-stabilization statement is dimension four, where the same proposition requires excluding points, curves *and surfaces*. Everything else — the projective-bundle formula putting two copies of $\alpha(X)$ in $X \times \mathbf{P}^1$, the non-rationality criterion, and the decoration itself — is already theirs. So the endpoint theorem may reduce to one classification question, which in the terms of this memo reads:

> **(iii)** Every atom of a smooth projective surface has formal monodromy of order at most two, its eigenvalues lying in $\{+1, -1\}$. In particular no such atom has a primitive sixth root of unity as an eigenvalue, and no Serre operator on such an atom's numerical Grothendieck group has $\Phi_6$ as a factor.

The stronger form is the one Katzarkov–Kontsevich–Pantev–Yu's own dimension-four example uses, where the exclusion is unipotency against non-unipotency. Their Example 6.20 needs it only for surfaces of general type, the atom there being shaped like one; the endpoint needs every surface, because the cubic threefold's atom is shaped like a genus-five curve atom and nothing narrows the search in advance.

Two reductions come with the statement, and one warning.

*Rank-one atoms are automatic.* For a rank-one lattice the Euler form is a $1 \times 1$ matrix, hence symmetric, so $S = E^{-1}E^T = 1$. A semisimple small quantum cohomology has every block of rank one, so (iii) holds for every surface with semisimple small quantum cohomology with nothing to prove. For a surface with nef canonical class their Lemma 5.24 gives a single atom and Claim 6.15 gives a regular singularity with nilpotent residue after the half-parity gauge, putting every eigenvalue at a root of unity of order at most two.

*The remaining surfaces, by the minimal model program.* These two reductions leave a short induction, which closes (iii) modulo three imported statements.

**Proposition 27** (Step (iii) for surfaces)**.** *Assume Katzarkov–Kontsevich–Pantev–Yu's Lemma 5.24 and Claim 6.15. Then every atom of a smooth projective surface has formal monodromy of order at most two; in particular none has a primitive sixth root of unity as an eigenvalue. The same holds for Serre eigenvalues, either because their Serre automorphism is defined as the monodromy in the u-direction, or case by case from the Serre computations recorded alongside each step of the proof.*

*Proof.* Induct on the number of blow-downs to a minimal model, finite by Castelnuovo, with the statement quantified over all parameters of the formal base. If $S$ is not minimal, write $S = \mathrm{Bl}_p S'$. Iritani's decomposition theorem (arXiv:2307.13555, Theorem 5.18) gives an isomorphism of quantum $D$-modules $\mathrm{QDM}(S) \to \tau^*\mathrm{QDM}(S') \oplus \varsigma^*\mathrm{QDM}(\mathrm{pt})$ commuting with the quantum connection and intertwining the pairings, and by his Corollary 1.2 the underlying change of variables preserves the Euler vector fields. So the formal type of $S$ at a parameter is that of $S'$ at the corresponding parameter plus that of a point, which is rank one with trivial connection; the added atom contributes the eigenvalue 1 and $S$ inherits the conclusion from $S'$. A minimal surface has nef canonical class, or is $\mathbf{P}^2$, or is a projective bundle over a smooth curve. The nef case is the reduction above. For $\mathbf{P}^2$ the characteristic polynomial of $c_1\star$ is $\lambda^3 - 27q$, of nonzero discriminant, so the small quantum cohomology is semisimple and every atom is rank one, so Theorem 13 applies to each. For a projective bundle the atoms are copies of the base curve's, and the curve is settled on the monodromy side directly, with no appeal to the identification. Let $C$ have genus $g$. The even part of $H^*(C)$ is spanned by 1 and the point class, and the grading operator has eigenvalues $\mp\frac{1}{2}$ on them. If $g \geq 1$ there are no rational curves and $\deg(-K_C) \leq 0$, so the quantum product is the cup product and $E\star$ is nilpotent: the even part is a single block at the eigenvalue 0, its connection is exactly $z^{-1}E\star - \mu$ with no higher $z$ coefficients, and the shearing moves the exponents $\pm\frac{1}{2}$ by integers only, which $\nu_6$ does not see. The formal monodromy eigenvalues are therefore $e^{\mp\pi i} = -1$, twice. If $g = 0$ then $E\star = 2H\star$ has the distinct eigenvalues $\pm 2\sqrt{q}$, so both blocks are multiplicity one and Theorem 13 applies again. Either way no primitive sixth root. The Serre side agrees, as it must if the identification holds: the Euler form $\left(\begin{smallmatrix} 1-g & 1 \\ -1 & 0 \end{smallmatrix}\right)$ on $([\mathcal{O}_C], [\mathcal{O}_{pt}])$ gives the Serre operator $\left(\begin{smallmatrix} -1 & 0 \\ 2-2g & -1 \end{smallmatrix}\right)$, of characteristic polynomial $(\lambda+1)^2$ for every $g$. This exhausts the classification. $\square$

The fake of the previous paragraph does not disturb the blow-down step: it mixes the new exceptional object with the ambient structure sheaf, which lie in different atoms, so the pair it generates is not an atom. Nor is the blow-down step resting on a multiplicity count: the computational literature — Böhning, von Bothmer and Su'a — verifies the blowup statement for multiplicity sequences and records that naive and small atomic decompositions can differ, but Iritani's theorem is at the level of the connection itself, which is the reading used above. The reading list, cache locators and hashes are in '2026-08-15-c912-blowup-formula-source-check.md'. Iritani also records that Gyenge and Szabó studied the asymptotics of the Euler eigenvalues under blowdowns to minimal models of surfaces, which is the configuration this induction runs through.

*The ambient category never sees it.* The Serre functor of a surface is tensor by $\omega$ followed by $[2]$, and $[2]$ acts by $+1$ on K-groups, so the ambient numerical Serre operator is multiplication by $\exp(K)$, which is unipotent. A primitive sixth eigenvalue on a surface can only come from a decomposition, never from the surface's own K-theory.

*The warning: admissible is not enough.* For a subcategory generated by an exceptional pair with $\chi(e_1, e_2) = x$ the numerical Serre operator has characteristic polynomial $\lambda^2 + (x^2 - 2)\lambda + 1$, which is $\Phi_6$ exactly when $x^2 = 1$. On any surface carrying a $(-1)$-curve $E$ — that is, any non-minimal surface — the sheaf $O_E(-1)$ is exceptional, $\mathrm{Hom}^\bullet(O_X, O_E(-1)) = 0$, and $\chi(O_E(-1), O_X) = -1$, using only $E^2 = -1$ and $K_X \cdot E = -1$. So $\langle O_E(-1), O_X \rangle$ generates an admissible subcategory whose Serre operator has order six. Statement (iii) is therefore false if "atom" is weakened to "admissible subcategory", and no lattice argument ranging over admissible subcategories can prove it. The subcategory is not an atom: it is generated by an exceptional pair and splits into two rank-one pieces whose Serre operators are both the identity.

Since the blowup is exactly what supplies this fake, the blowup step of any proof of (iii) is where the care is needed.

With all four imports now read at the source, the route assembles: Theorem 4.11 with $r = 2$ puts two copies of the cubic's atom in $X \times \mathbf{P}^1$; that atom has primitive-sixth monodromy by the exponents $-1/6, -5/6$ above; no atom of a variety of dimension at most two has such an eigenvalue, by Proposition 27 for surfaces together with the point and curve cases; so Proposition 5.30 in dimension four gives that $X \times \mathbf{P}^1$ is not rational. This uses no transport lemma and none of Section 9.

This is not the same invariant as $\nu_6$, and it is not obviously weaker or stronger; it is the sources' own, which is an argument for using it. Two cautions. It buys the Hodge-atom, motive and Serre-enhancement machinery that the earlier blueprint deliberately avoided, so the manuscript would import more than the two decomposition theorems — though not the integral-structure enhancement, which its authors call difficult and defer to forthcoming work, while calling the Euler-pairing and Serre-automorphism enhancement a straightforward repeat of the undecorated theory. And their sentence is a worked example, not a theorem with hypotheses, although the statements it invokes — Lemma 5.24, Claim 6.15, Proposition 5.30 and Theorem 4.11 — have now been read at the source and say what is claimed of them here. The graded minimal polynomial has been checked on the lattice as well, and the two conventions differ as described above; the surface exclusion is Proposition 27. Backing computations, and the exact record of what was and was not read at the source, are in '2026-08-15-c912-step-iii-serre-restatement.md' and '2026-08-15-c912-step-iii-surface-induction.md', with the scripts '2026-08-15-c912-surface-atom-serre-check.py' and '2026-08-15-c912-step-iii-surface-induction-check.py'.

## Arbitrary Jordan size: what extends, and the one thing that does not

Earlier versions of this section closed with the expectation that the mechanism — commutant of a regular block kills the would-be $z^{-2}$ bulk term, one shearing makes the block regular singular, flatness then forces the bulk pole to vanish — should extend to a single Jordan block of any size with the shearing $\operatorname{diag}(1, z, \ldots, z^{m-1})$, since that is what a general center statement needs. Carrying it out separates cleanly into a part that does extend, a bookkeeping statement, and one step where rank two turns out to be genuinely special.

**Proposition 28** (Steps 1 to 3 in size $m$)**.** *Let $H_0$ be a generalized eigenspace of $U(0)$ of rank $m$ on which the nilpotent part is regular, the other eigenvalues differing from $u_0$ by units. Put $u_0 = \frac{1}{m}\operatorname{tr}(U|_{H_0})$ and $N := U|_{H_0} - u_0 I$. Then*

1. *$C_{a,0} \in B[N] = B \cdot I \oplus B \cdot N \oplus \cdots \oplus B \cdot N^{m-1}$, and the coefficient of $I$ is $\partial_a u_0$; after the twist by $\exp(u_0/z)$ the leading bulk coefficient lies in $N \cdot B[N]$, hence is strictly upper triangular in a cyclic frame, so no entry of it shears down to a pole of the bulk connection;*

2. *$\operatorname{tr}(N^k) \equiv 0$ on $B$ for every $k \geq 1$. Hence $N$ is nilpotent at every point of the formal germ, and regular there because $N^{m-1}$ has a unit entry, so the block keeps one eigenvalue of multiplicity $m$ and a single Jordan block.*

*Proof.* $[U, C_a] = 0$ gives $[U|_{H_0}, C_{a,0}] = 0$, and the commutant of a regular matrix over a local ring is the free module it generates, which is $B[N]$. Write $C_{a,0} = p_a I + Q_a$ with $Q_a := \sum_{j \geq 1} q_a^{(j)} N^j \in N \cdot B[N]$; this much is the commutant statement alone, and the identification of $p_a$ is deferred to the end.

(2) By Theorem 11, $D_aN = Q_a + [Q_a, \mu_0]$. Put $t_k := \mathrm{tr}(N^k)$, so $t_1 \equiv 0$ by construction. Then

$$\partial_a t_k = k\,\mathrm{tr}\big(N^{k-1}D_aN\big) = k\,\mathrm{tr}\big(N^{k-1}Q_a\big) = k\sum_{j\geq 1} q_a^{(j)} t_{k+j-1},$$

the commutator term dropping because $N^{k-1}$ and $Q_a$ commute and the trace is cyclic: $\mathrm{tr}(N^{k-1}Q_a\mu_0) = \mathrm{tr}(N^{k-1}\mu_0 Q_a)$. By Newton's identities the coefficients of the characteristic polynomial of $N$ are polynomials without constant term in $t_1, \dots, t_m$, so Cayley–Hamilton puts every $t_l$ with $l > m$ in the ideal generated by $t_2, \dots, t_m$. The vector $t = (t_2, \dots, t_m)$ therefore satisfies a linear system $\partial_a t = M_a t$ with $M_a$ over $B$, and $t(0) = 0$ because $N(0)$ is nilpotent. The lowest-order argument of Theorem 18 applies to the vector unchanged, so $t \equiv 0$, and in characteristic zero the vanishing of all power traces forces the characteristic polynomial of $N$ to be $\lambda^m$.

(1) With $\mathrm{tr}\,N^j \equiv 0$ for every $j \geq 1$ now available, $\mathrm{tr}\,C_{a,0} = m\,p_a$, and taking traces in $\partial_a U = C_a + [C_a, \mu]$ on the block gives $m\,\partial_a u_0 = m\,p_a$. The twist by $\exp(u_0/z)$ replaces $C_{a,0}$ by $C_{a,0} - (\partial_a u_0)I = Q_a$, which lies in $N \cdot B[N]$ and so has no entry on or below the diagonal. $\square$

The shearing bookkeeping is the next thing to fix, because it decides what has to be proved. Conjugation by $S = \mathrm{diag}(1, z, \dots, z^{m-1})$ multiplies the $(i,j)$ entry by $z^{j-i}$, so writing the twisted decoupled block connection as $z^{-1}N + A'_0 + zA'_1 + \cdots$, the coefficient of $z^{-k}$ in the sheared connection collects the entries of $A'_p$ at sub-diagonal distance $i - j = p + k$, and $z^{-1}N$ shears up to the residue. Hence:

> the sheared block is regular singular exactly when $A'_p$ vanishes strictly below its $p$-th sub-diagonal, for every $p \geq 0$.

For $m = 2$ this is the single condition $f = (A'_0)_{21} = 0$. For $m \geq 3$ it is a nested family, deepest first: $(A'_0)_{m1}$, then $(A'_0)_{m-1,1}$, $(A'_0)_{m,2}$ and $(A'_1)_{m,1}$, and so on.

**Proposition 29** (Duality in size $m$, and its limit)**.** *In the cyclic frame $e_i = N^{m-i}e_m$ the Gram matrix of $G_0$ is anti-triangular Hankel: $(e_i, e_j)$ depends only on $i+j$ and vanishes for $i+j \leq m$, so $G_0 = J\,P(N)$ with $J$ the reversal matrix and $P(N) \in B[N]$ invertible. Writing $X^{\mathrm{at}} := JX^TJ$ for the reflection in the anti-diagonal, the parity rule (8.5) becomes*

$$(A'_p)^{\mathrm{at}} = (-1)^{p+1}\,P\,A'_p\,P^{-1}.$$

*The reflection preserves each sub-diagonal, and conjugation by $P$ acts trivially on the deepest one, so $(A'_p)_{m1} = 0$ for $p$ even and is unconstrained for $p$ odd; on the next sub-diagonal the rule only exchanges the two entries with each other.*

*Proof.* Self-adjointness of $N$ gives $(e_i, e_j) = (N^{m-i}e_m, N^{m-j}e_m) = (e_m, N^{2m-i-j}e_m)$, which depends only on $i+j$ and vanishes when $2m-i-j \geq m$; the anti-diagonal value $(e_m, N^{m-1}e_m)$ is a unit by nondegeneracy. Collecting anti-diagonals gives $G_0 = \sum_{t\geq 0} g_{m+1+t}JN^t = J\,P(N)$. Substituting into $(A'_p)^TG_0 = (-1)^{p+1}G_0A'_p$ and multiplying by $J$ on the left gives the displayed rule, using $J^2 = I$. The reflection sends $(i,j)$ to $(m+1-j, m+1-i)$ and so preserves $i-j$; and in $PA'_pP^{-1}$ every correction term is of the form $N^aA'_pN^b$ with $a+b \geq 1$, whose $(m,1)$ entry is $(A'_p)_{m+a,1-b}$ and therefore out of range. At distance $m-1$ the rule thus reads $(A'_p)_{m1} = (-1)^{p+1}(A'_p)_{m1}$. $\square$

So duality removes exactly one pole coefficient, the deepest, $z^{-(m-1)}$. In rank two that coefficient is the entire pole, which is why Theorem 20 closes the case; for $m \geq 3$ it is one condition out of a growing family. The expectation that the rank-two mechanism extends as it stands was therefore too optimistic at exactly this step, and the extension needs a different input at the base point.

The natural input is the grading rather than the pairing, and it is available precisely for the blocks a center statement cares about, those at $u_0 = 0$. Since $\mu$ is the grading operator and $E\star$ raises degree by two, $[\mu, U] = U$; this makes $H_0$ for the eigenvalue zero $\mu$-invariant, and on it $[\mu, N] = N$, so $N$ raises the $\mu$-weight by one. In a $\mu$-homogeneous cyclic frame an operator of $\mu$-weight $p$ is supported exactly on the $p$-th sub-diagonal, so the statement "$A'_p$ has $\mu$-weight $p$" *is* the regular-singularity condition displayed above, with nothing left to check entry by entry. The cubic is the rank-two instance of it: $A'_0 = D_0$ is diagonal, of weight zero, and $A'_1$ contributes only its $(2, 1)$ entry, of weight one — which is why the deleted hypothesis (H2) looked like an accident of that example and was not one.

What must be supplied to turn this into a proof is that the decoupling gauge of Step 1 respects the grading on the zero block: the analogue of Lemma 19 with $\mu$ in place of $G$. Uniqueness is again the device, but the argument is not verbatim, because the $\mathbf{C}^\times$ action generated by $\mu$ scales the nonzero eigenvalues of $U$ rather than fixing them, so it moves the other blocks and the equivariance has to be stated for the whole decoupling problem before restricting to $H_0$. With the base point settled that way, propagation over the germ should be Theorem 21's flatness argument applied to the pole coefficients in order of depth; the shape to expect is that each satisfies a linear equation whose remaining terms involve only deeper coefficients, already known to vanish, but that triangular structure is a computation this memo has not done.

None of this covers a derogatory block, where the commutant is larger than $B[N]$ and Proposition 28 fails at its first step, nor a semisimple block, where the $z^{-1}$ equation instead reads $(I + \mathrm{ad}_R)(C_{a,0} - p_a I) = 0$ and a resonance $\rho_i - \rho_j = -1$ must be excluded before the same conclusion follows.

## 8 The one-stabilization theorem, and the one step it still needs

Section 7 closes the residual gap for the cubic block, but the draft's own presentation needs the same statement at every intermediate fourfold of a weak factorization, and that is a statement about arbitrary blocks. This section changes where the bookkeeping happens. In the atom formulation the intermediate fourfolds never appear, because the weak-factorization ledger is an identity in the free abelian group on atoms. That removes the transport problem completely. It introduces one requirement of its own, which the first draft of this section wrongly called standard: the invariant must be well defined on atoms, and an atom is a geometric atomic $F$-bundle modulo isomorphism *and* modulo change of base point within a connected component of the spectral cover. The second equivalence is where the argument is still open, and Section 8 states exactly why.

### The invariant

For a smooth projective $T$ and a point $b$ of the even base of its A-model $F$-bundle, let $\mathrm{Mon}(T, b)$ be the multiset of eigenvalues of the framed monodromy after Levelt–Turrittin separation, and

$$\nu_6(T, b) := \#\{\zeta \in \mathrm{Mon}(T, b) : \zeta = e^{\pi i/3} \text{ or } e^{-\pi i/3}\},$$

with algebraic multiplicity. The same definitions apply to a summand of the spectral decomposition, in particular to a geometric atomic $F$-bundle. At a point of the base the coefficient field is $\Lambda(\!(z)\!)$, so Levelt–Turrittin is available and Section 2 applies with $\Gamma = 0$: no receiver enters anywhere in this section.

## Bulk local constancy

**Theorem 30** (The formal type is constant on the base)**.** *Let $(H, \nabla)$ be the A-model $F$-bundle of a smooth projective $T$ over its base $B$, and let $b \in B$. Then the framed monodromy at every point of the formal germ of $B$ at $b$ is conjugate to that at $b$. Consequently $\nu_6$ is locally constant on $B$, hence constant on each connected component, and likewise on each summand of the spectral decomposition.*

*Proof.* This is Cai's argument in the proof of his Proposition 6, which deserves to be stated in this generality. Write $\nabla_{t_i} = \partial_{t_i} + z^{-1}P_i$ for the bulk directions, with $P_i$ quantum multiplication by $\phi_i$, and solve

$$\partial_{t_i} M = -z^{-1} P_i M, \qquad M|_{t=0} = I$$

recursively for $M = I + \sum_n M_n t^n$. The recursion produces $M_n \in \mathrm{End}(H) \otimes \Lambda[z, z^{-1}]$, with $z$-order bounded below by $-|n|$ at each fixed bulk degree, so $M \in \mathrm{GL}\big(\mathrm{End}(H) \otimes \Lambda[z, z^{-1}][[t]]\big)$; invertibility is automatic from $M = I + O(t)$. By construction $M^{-1}\nabla_{t_i} M = \partial_{t_i}$, and since the $\nabla_{t_i}$ commute with $\nabla_z$, so do their conjugates; hence $M^{-1}\nabla_z M = \partial_z + L$ with $\partial_{t_i} L = 0$, and $L$ is the $z$-connection at $b$ because $M|_{t=0} = I$.

Let $Y$ be a formal fundamental solution at $b$, possibly after a ramification $z = w^e$, with $\sigma(Y) = Y M_{\mathrm{f}}$ for the turn $\sigma$. Then $MY$ is a fundamental solution at the deformed parameter. Every entry of $M$ is a Laurent series in *integral* powers of $z$, so $\sigma(M) = M$ by Lemma 3, and

$$\sigma(MY) = M\sigma(Y) = (MY)M_{\mathrm{f}}.$$

The framed monodromy is therefore the same matrix in this solution frame, and well defined up to conjugacy over the constants by Proposition 6. A locally constant integer-valued function on a connected base is constant. □

## The gap, stated exactly

Theorem 30 is a statement about the *formal* germ: the deformed connection is gauge equivalent to the one at $b$ over $\Lambda[z, z^{-1}][[t]]$, with $t$ formal. The atom equivalence asks for more. Two geometric atomic $F$-bundles of the same variety are identified when their spectral lifts lie in the same connected component, and two points of one connected component of a non-archimedean analytic space are not in general related by a chain of formal germs. What is needed is local constancy in the analytic topology, and then connectedness.

The obstruction to upgrading is not bookkeeping. It is the same divergence this memo has been circling from the start. The gauge $M$ of Theorem 30 has $z$-order bounded below by $-|n|$ in bulk degree $n$, so it converges when the bulk parameter is topologically nilpotent and not otherwise; at a bulk parameter that is merely nearby in the analytic topology, the sum over bulk degrees has no lower bound in the loop coordinate, and $M$ is not a gauge in any ring where the turn argument runs. So the receiver problem of Sections 2 and 3 is not escaped by moving to atoms; it is relocated to the globalization step.

One route to closing it is visible, and it is Section 7 rather than Cai's gauge. That argument produces the differential identity $\partial_a R = [R, G_a]$ between honest matrix-valued functions on the base, and a function with identically vanishing derivative on a smooth connected space in characteristic zero is locally constant. So the shearing argument globalizes where the gauge argument does not, and what it needs in exchange is its two hypotheses at every point of the component: the block must stay a rank-two Jordan block, and it must stay regular singular after one shearing. Theorem 18 propagates nilpotency and Theorem 21 propagates regularity, but neither excludes the locus where the nilpotent part degenerates to zero, which is where the argument would have to be extended.

*Remark* 31 (What the gap does and does not touch). It does not touch Theorem 34: the nef-canonical input there is pointwise. It does not touch the computation $\nu_6(\alpha(X)) = 2$ at the hyperplane point, nor its stability under specialization, which is Corollary 24. It touches exactly the claim that these two facts may be compared as statements about one atom.

*Remark* 32 (Why this is not in tension with the receiver problem). The gauge $M$ is the pro-Laurent object of Section 5, and the whole receiver discussion of Sections 2 and 3 was about *specializing* the bulk parameter to an element with Novikov coefficients, where the sum over bulk degrees loses its lower bound in $z$. No such specialization occurs here: Theorem 30 compares two points of one base along a formal germ, where the bulk parameters are formal variables and $M$ has Laurent coefficients degree by degree. That is exactly the difference between the two formulations, and it is why the atom route is free of the criterion $e\,w(\Delta\lambda) < \varepsilon$. Section 7 remains the stronger statement where it applies, since it produces a characteristic polynomial constant in $\tau$ and so survives specialization; Theorem 30 is the general one.

## The invariant descends to atoms

**Proposition 33.** *$\nu_6$ is invariant under isomorphism of geometric atomic $F$-bundles. It is well defined on atoms if and only if it is in addition constant on connected components of the spectral cover of each variety, which is open.*

*Proof.* Atoms are geometric atomic $F$-bundles modulo the equivalence generated by isomorphism of atomic $F$-bundles and by change of the base point within one connected component of the spectral cover of one variety. An isomorphism carries the connection, hence the Levelt–Turrittin separation and the framed monodromy, so it preserves Mon and $\nu_6$. The second equivalence is exactly the constancy statement, which Theorem 30 supplies along formal germs only. □

It is worth being precise about why the weaker statement does not suffice. One might hope to compare the set of values of $\nu_6$ over the representatives of a class, which is well defined without any constancy. It does not help: the class one is trying to exclude is precisely a class whose value set would contain both 2, from the cubic representative, and 0, from a surface representative, and a non-constant value set is not by itself absurd. The contradiction needs constancy, not a value set.

## No atom in dimension at most two carries the invariant

**Theorem 34.** *Let $\alpha$ be an atom represented by a smooth projective complex variety of dimension at most two. Then $\nu_6(\alpha) = 0$.*

*Proof.* Induction over the classification, using the two chemical-formula identities of Katzarkov–Kontsevich–Pantev–Yu's Proposition 5.22, namely $\mathrm{CFF}(\mathbf{P}(V)) = \mathrm{rank}(V)\,\mathrm{CFF}(T)$ and $\mathrm{CFF}(\mathrm{Bl}_Z T) = \mathrm{CFF}(T) + (\mathrm{codim}\, Z - 1)\,\mathrm{CFF}(Z)$, both of which follow from their Theorems 4.5 and 4.11.

A point has a one-dimensional cohomology with trivial connection, so its atom has trivial framed monodromy and $\nu_6 = 0$.

If $K_T$ is nef, their Lemma 5.24 gives a single atom, and their Claim 6.15 gives a gauge, by $u^g$ with $g = \mathrm{Gr} + \frac{1}{2}T_{\mathrm{d}}$, after which the connection has a first-order pole with nilpotent residue. Undoing that gauge shifts the exponents by the eigenvalues of $g$, which lie in $\frac{1}{2}\mathbf{Z}$. So the framed monodromy is unipotent on even cohomology and has eigenvalue $-1$ on odd cohomology, and $\nu_6 = 0$. This input is pointwise and needs no constancy statement: their proof shows that for $K_T$ nef, at any even base point with vanishing $H^0$-coordinate, quantum multiplication by the Euler field moves the grading eigenspace for $m$ into eigenspaces for values at least $m + 1$, so its block components $\kappa_{ij}$ vanish for $j \leq 0$ and the gauged operator has a first-order pole. The normalization of the $H^0$-coordinate costs nothing, since translation in the identity direction only twists by a scalar exponential.

Curves: $\mathbf{P}^1 = \mathbf{P}(\mathcal{O}^{\oplus 2})$ over a point, so its atoms are point atoms; a curve of genus at least one has nef canonical class.

Surfaces: every smooth projective surface is obtained from a minimal one by blowing up points, and a point has $\nu_6 = 0$, so by the blowup identity it suffices to treat minimal surfaces. A minimal surface of Kodaira dimension at least zero has nef canonical class. A minimal surface of Kodaira dimension $-\infty$ is $\mathbf{P}^2 = \mathbf{P}(\mathcal{O}^{\oplus 3})$ over a point, or a geometrically ruled surface $\mathbf{P}(V)$ over a curve; in the first case the atoms are point atoms, in the second they are curve atoms.

In every case the atoms are among those already shown to have $\nu_6 = 0$. $\square$

Note that this argument is entirely at the level of atoms as classes, so no bulk point is chosen and no transport is used; the only pointwise input is Claim 6.15, which is a structural statement about a connection with nef canonical class.

## The theorem

**Theorem 35** (One stabilization)**.** *Assume $\nu_6$ is constant on connected components of the spectral cover, as discussed in Section 8. Let $X \subset \mathbf{P}^4_{\mathbf{C}}$ be a smooth cubic threefold. Then $X \times \mathbf{P}^1$ is irrational.*

*Proof.* At the hyperplane point of the base of $X$, the spectral decomposition has two one-dimensional summands for the nonzero eigenvalues of the Euler operator and one summand $\alpha(X)$ for the eigenvalue 0; this is Katzarkov–Kontsevich–Pantev–Yu's Example 6.21, and it is the draft's own $K_0$ with eigenvalues $\pm 6r, 0, 0$. The draft's computation, which is Cai's Proposition 6, puts formal solutions with $z^\rho$, $\rho \equiv \pm 1/6$, in the rank-two block for the eigenvalue 0; equivalently the sheared residue of Section 7 has characteristic polynomial $\rho^2 + \rho + 5/36$. Hence $\nu_6(\alpha(X)) \geq 2$, and it is well defined by Proposition 33.

Since $X \times \mathbf{P}^1 = \mathbf{P}_X(\mathcal{O}_X^{\oplus 2})$, the projective-bundle identity gives $\mathrm{CFF}(X \times \mathbf{P}^1) = 2\,\mathrm{CFF}(X)$, so $\alpha(X)$ occurs in the atomic composition of $X \times \mathbf{P}^1$.

Suppose $X \times \mathbf{P}^1$ were rational. By the non-rationality criterion, their Proposition 5.17 in the equivariant form and Proposition 5.30 in the Hodge form, every atom in its atomic composition would be represented by a smooth projective variety of dimension at most $4 - 2 = 2$.

Then $\alpha(X)$ would be such an atom, and Theorem 34 would give $\nu_6(\alpha(X)) = 0$, contradicting $\nu_6(\alpha(X)) \geq 2$. $\square$

The invariant does not need to be computed exactly, and the odd cohomology $H^3(X)$ may well lie in the same summand; only the presence of the two primitive-sixth eigenvalues is used, and they cannot be cancelled because $\nu_6$ counts with multiplicity in a free abelian group.

## What is imported, and what is proved here

Imported, and not reproved:

- The blowup decomposition of maximal $F$-bundles, and the projective-bundle decomposition. These are the theorems of Iritani and of Iritani–Koto, packaged as Theorems 4.5 and 4.11; reproving them would be far larger than this theorem, and no presentation should claim independence of them.
- The chemical-formula identities and the non-rationality criterion, which are formal consequences of those two theorems together with weak factorization.
- Claim 6.15 for a variety with nef canonical class, and Lemma 5.24.
- Weak factorization in characteristic zero.

Proved here or in the draft: constancy of the formal type along formal germs (Theorem 30), invariance of $\nu_6$ under isomorphism (Proposition 33), the low-dimensional vanishing (Theorem 34), the cubic block computation, and the specialization-safe rigidity of Section 7.

Open: constancy of $\nu_6$ on connected components, which is Section 8. Hypothesis 4.7H does not appear anywhere in the chain — no framed operator is transported across a comparison isomorphism — so the residual conditionality is of a different kind and a smaller one, but it is conditionality and the theorem should be stated with it until it is removed.

## Which other routes also close

Three, at different costs.

*The draft's own presentation.* Keep the operation formulas and transport $\nu_6$ summand by summand. This needs the specialization-safe statement at every intermediate fourfold, so it needs Section 7 generalized from a rank-two Jordan block to an arbitrary block. That generalization is plausible for a single Jordan block of any size by the same shearing, and needs a resonance hypothesis in the semisimple case; it is real work, and it is the only route that keeps the draft's current architecture.

*The sources' Serre decoration, which is now the leading candidate.* It avoids the gap of Section 8 by construction, and this is the strongest reason to prefer it. Their Proposition 5.23 proves that the isomorphism class of the fibre representation of an atom is independent of the representative, using rigidity of finite-dimensional representations of a proreductive group — a global argument, valid across a connected component, which monodromy data does not enjoy. They then decorate with a Serre automorphism and state that the enhanced theory is a straightforward repeat of the undecorated one, with the corresponding strengthened non-rationality criterion.

Their Example 6.21 separates $\alpha(X)$ from a genus-five curve atom by the graded minimal polynomial $S^5 = [3]$ and concludes that every smooth cubic threefold is irrational. More to the point for the endpoint, their Example 6.17 runs the same pattern in *dimension four*, for a very general cubic fourfold containing a plane: there $\alpha(X)$ is shaped like the atom of a surface of general type, and they exclude it by playing $S^3 = [4]$ against the unipotency that Claim 6.15 forces on the even cohomology of a nef-canonical surface. That is exactly the shape the one-stabilization argument needs, with $X \times \mathbf{P}^1$ in place of their fourfold, and it pairs their globally well-defined decoration on the cubic side with a pointwise structural statement on the surface side — the two halves whose mismatch is the gap above.

Two cautions remain. The enhanced theory is asserted straightforward rather than written out, and the integral-structure enhancement is explicitly deferred to forthcoming work; only the Euler-pairing and Serre-automorphism enhancements are claimed here. And their examples are worked examples, not theorems with hypotheses, so the surface exclusion for $X \times \mathbf{P}^1$ has to be done rather than cited.

*Both decorations at once.* They are independent invariants of the same atom, so proving the surface exclusion twice is a genuine referee-facing check rather than duplication. If the Serre computation is cheap it is worth having as a second column in the same table.

# 9 The endpoint: what is proved, and the one statement left

The first version of this section claimed the endpoint unconditionally by inverting Iritani's change of variables. That was refuted, and the refutation is recorded here because the move is tempting and should not be made again. Iritani–Koto's (5.13) sets $s_j = \varsigma_j - \varsigma_j^\circ$ and states that the invertible change of variables is the one in those *displaced* coordinates, treated as independent formal variables; Iritani's Lemma 5.15 is the formal inverse function theorem at $Q = \vartheta = 0$, an isomorphism of the germ at the origin onto the germ at its image; and Iritani states that the pullback of functions along the change of variables is ill-defined. Every version of the decomposition identity in the sources is an isomorphism of modules over a formal germ, not a family of pointwise statements. So a parameter at which a prescribed displaced coordinate vanishes need not exist, and "read the identity in either direction" is false.

What replaces it is better, and it splits the endpoint cleanly into a computed half and one remaining statement.

### The displacement for a trivial bundle is a pure string shift

**Theorem 36.** *Let $V = \mathcal{O}_X^{\oplus 2}$. Then the Iritani–Koto initial displacement is $\varsigma_j^\circ = r\lambda_j = \pm 2q^{1/2}$ exactly, a class in $H^0$, with no tail.*

Two independent proofs, which is what the claim deserves given that an earlier version of this memo asserted the opposite.

First, from their (5.11), $\varsigma_j^\circ = r\lambda_j + [z^{-1}] \log\big(q^{c_1(V)/(rz)} F_j(1)\big)$. For a trivial bundle $c_1(V) = 0$, so the exponential prefactor is absent, and all Chern roots vanish, so $F_j(1)$ is a scalar series. A power count in the stationary-phase transform then shows it has no negative powers of $z$: a factor from $\exp(-\varphi_{\geq 3}/z)$ contributes $z^{-a}$ with $s$-degree at least $3a$, the remaining tail contributes $z^{+b}$ with $b \geq 0$, and Gaussian pairing of total $s$-degree $m$ contributes $z^{m/2}$, so the net power is at least $a/2 + b \geq 0$. The $z^{-1}$ coefficient is therefore zero. The $O(q^{-1/r})$ that appears in the sources is a bound on $F_j(1)$, not a nonvanishing statement about $\varsigma_j^\circ$; conflating the two is what the earlier version did.

Second, and needing none of that: $\mathbf{P}(\mathcal{O}_X^{\oplus 2}) = X \times \mathbf{P}^1$, and the small quantum cohomology of a product is the tensor product of the factors'. So $E\star = E_X \star \otimes 1 + 1 \otimes E_{\mathbf{P}^1}\star$, and the generalized eigenspace for the cluster at $r\lambda_j$ is $H^*(X) \otimes v_j$ for $v_j$ the $j$-th eigenline of $E_{\mathbf{P}^1}\star$. The grading operator compresses to $\mu_X + \operatorname{tr}(P_j \mu_{\mathbf{P}^1})$, and those numbers are degree zero, permuted by the deck action $q^{1/r} \mapsto e^{2\pi i/r} q^{1/r}$, hence equal, and they sum to $\operatorname{tr}\mu = 0$, hence vanish. The $j$-th summand is $z^2\partial_z = (U_X + r\lambda_j) - z\mu_X$, which is the cubic's quantum connection at the bulk parameter $r\lambda_j \cdot 1 \in H^0$. Since generalized eigenspaces are canonical rather than chosen, this pins $\varsigma_j^\circ = r\lambda_j$ with no appeal to the transform at all.

**Corollary 37.** $\nu_6(X \times \mathbf{P}^1, 0) = 4$*, as a computed fact at a parameter that indisputably exists.*

*Proof.* An $H^0$ shift changes the connection by $(\tau^0/z) \cdot \mathrm{id}$, which shifts every exponential factor uniformly and changes no block's regular part. This is exact and needs no smallness, which matters because $\pm 2q^{1/2}$ is of unit order. So each summand contributes the cubic's value at the origin, which is 2 by the block computation of Section 7. □

### The remaining statement, and why it is small

Weak factorization can be arranged as a roof $X \times \mathbf{P}^1 \leftarrow \cdots \leftarrow V \rightarrow \cdots \rightarrow \mathbf{P}^4$ with every arrow a blowdown. Descending from the roof is then a composition of *forward* substitutions only, so no inverse is evaluated anywhere and the refuted move is not needed. The centre terms vanish by the low-dimensional lemma, giving $\nu_6(X \times \mathbf{P}^1, \widehat{\tau}_L) = \nu_6(\mathbf{P}^4, \rho_R) = 0$ for the two parameters the two descents produce.

Against that stands Corollary 37. So the endpoint reduces to exactly one statement:

$$\nu_6(X \times \mathbf{P}^1, \widehat{\tau}_L) = \nu_6(X \times \mathbf{P}^1, 0), \text{ or merely } \nu_6(X \times \mathbf{P}^1, \widehat{\tau}_L) \geq 1.$$

Three things make this much smaller than anything attempted above. It concerns *one* variety rather than arbitrary intermediate fourfolds. Only a lower bound is needed, so no control of the other blocks is required. And the blocks carrying $\nu_6$ on $X \times \mathbf{P}^1$ are two copies of the cubic's rank-two Jordan block, which are exactly the hypotheses of Section 7.

The engine to run is therefore Section 7 along the pencil $t\widehat{\tau}_L$, $t \in [0,1]$, and not Cai's gauge: Cai's gauge at $t = 1$ still has $z$-order unbounded below across Novikov degrees, so it is not a gauge in $\Lambda(\!(z)\!)$ and the receiver criterion returns. Section 7 produces the differential identity $\partial R = [R, G]$ between $z$-regular objects, so if the pencil's connection matrices are polynomial in $t$ per slot — which the dimension axiom gives, each insertion of an $H^{\geq 4}$ class costing at least two — the identity holds over $\Lambda[t]$ and $t = 1$ is inside the domain rather than at its edge.

### The trade between the two sides, and where it cancels

It is worth being explicit about what is being traded, because the accounting localizes the remaining debt more sharply than the narrative does.

*Vertical motion is free; horizontal motion is the whole debt.* Each step of a factorization gives an exact identity $\nu_6(\mathrm{Bl}_Z V, w) = \nu_6(V, \tau(w)) + \sum_j \nu_6(Z, \varsigma_j(w))$, and with $\dim Z \leq 2$ the centre terms vanish. So moving between varieties conserves the invariant exactly: nothing leaks into a centre and nothing is created. The only way the invariant can change anywhere in the ledger is by moving between two parameters of *one* variety. Every obligation this memo has chased is a horizontal one, and the vertical steps have never been in doubt.

*Consequence, in contrapositive form.* Suppose $X \times \mathbf{P}^1$ is rational. The left end carries 4 and the right end carries 0, and conservation across every vertical step then forces some valley of the zigzag to have a bulk-parameter-dependent primitive-sixth multiplicity. So the theorem is equivalent to: no smooth projective fourfold arising as a valley has a $\nu_6$-carrying block that degenerates along the displacement the chain produces. Stated this way the debt is a single negative statement about degeneration, not a positive statement about transport.

*The balance has slack, and the slack is three.* The left end supplies 4, being two copies of the cubic's rank-two block, and the contradiction needs only 1. So up to three of the four eigenvalues may be lost anywhere along the chain without harming the argument. What must be excluded is total collapse, not any loss — a weaker statement than constancy, and one that Section 7 overshoots, since it proves the characteristic polynomial exactly constant.

*Four places where the debt cancels outright.*

1. Any step whose accumulated displacement lies in $H^0 \oplus H^2$. The string and divisor equations are exact there, so such a step costs nothing. This is not a rare case: it is exactly what makes the endpoint free, where the displacement is $\pm 2q^{1/2}$ and purely $H^0$.

2. An adjacent blowup and blowdown of the same centre, where the composite parameter map is the identity and no comparison of two parameters arises.

3. Any valley whose Euler operator is semisimple, or more generally has only multiplicity-one eigenvalues in the relevant range. There Theorem 13 gives $\nu_6 = 0$ at every parameter unconditionally, so the two incoming parameters agree trivially and the valley is vacuous.

4. Every centre, by the low-dimensional lemma. Centres are where a leak would have to go, and they cannot receive one.

*What is left after the cancellations.* The debt sits only at valleys that simultaneously carry a nonzero $\nu_6$ and receive a displacement with a nonzero $H^{\geq 4}$ part. By the dimension axiom each such displacement contributes finitely much per Novikov degree, so the surviving obligation is finite data at finitely many places, and by the slack it need only be shown not to annihilate all four eigenvalues at once. That is the sharpest available statement of what remains, and it is what a next attempt should target rather than constancy in general.

## Owed lemmas

1. Per-slot polynomiality in $t$ along the pencil, from the dimension axiom, stated for the accumulated parameter and not only for one step.

2. Nonvanishing of $\nu$ along the pencil. Section 7 needs the nilpotent part nonzero, and over $\Lambda[t]$ that is nonvanishing of a polynomial with $\nu(0) = 2$, not an open condition one may assume.

3. Legitimacy of the descent's substitutions past the first step, where the base parameter is evaluated at the accumulated parameter rather than at the origin. The accumulated parameter has only $H^{\geq 4}$ components, so finitely many insertions survive per Novikov degree; this is the same finiteness as item 1 and should be proved once.

4. The low-dimensional lemma, restricted to the parameters that actually arise rather than stated for all of them, since the unrestricted version invokes the operation identity at arbitrary parameters and that is exactly what the refutation forbids.

## 10 Routes, in the order I would now try them

The framing route has been removed from this list by the checks of Section 5, the blockwise route has been carried to its conclusion for a block of the cubic's type in Section 7, and the endpoint theorem is proved in Section 8. So this list no longer ranks attacks on an open problem; it records what each remaining route would still buy, which is the general statement rather than the endpoint.

1. **Blockwise, avoiding transport altogether.** Developed in Section 6 and completed in Section 7 for a single Jordan block of rank two. What remains of this route is generalization, not repair: a single Jordan block of arbitrary size, by the same shearing with $\mathrm{diag}(1, z, \ldots, z^{m-1})$, and the semisimple case with its resonance condition. That generalization is what a statement covering arbitrary centers would need; it is not needed for the cubic endpoint. Signs follow the displayed connection of Iritani–Koto's Theorem 5.1 and should be rechecked against the draft's normalization before anything is written into the manuscript.

2. **Birkhoff.** The identity $(\Phi^{\circ})^{-1} \circ M = M' \circ \Phi^{-1}$ of Iritani–Koto's Section 5.8, and its counterpart in Iritani's Section 5.8.2, already factors the bulk transport into a $z$-regular factor, which Theorem 7 handles, and a factor $M' = \mathrm{id} + O(z^{-1})$. What must be shown is that the second factor does not change the primitive-sixth multiplicity. This is a statement about a $z = \infty$-side gauge acting on a $z = 0$ invariant, so it is not automatic, and it is close to the original difficulty in another form: any proof will have to use where $M'$ comes from, which is the flat quantum connection — that is, the identities route 1 already exploits directly. Worth keeping as a cross-check rather than as the primary attack.

3. **The low-dimensional centers directly**: primitive-sixth vanishing after Iritani's specialization, using the classification the draft already proves, rather than through a general transport statement.

4. **Divisor tagging on its own terms**: whether the divisor equation gives a more explicit transport than the general positive-bulk gauge of Lemma 4.5. Their Lemma 2.24 is the same mechanism in print and is worth citing; it does not close the lemma, for the reasons under check three.

5. **Normalize, then verify.** Replace "choose $L$ so large" by the smallest weight satisfying separation, adopt the single-copy receiver of Section 3, and verify $e\,w(\Delta\lambda) < \varepsilon$ case by case. This is fallback machinery now.

The iterated receiver of Section 4 is not on the list: it needs a semipositivity hypothesis on the ambient variety that weak factorization does not supply. It remains worth having where it applies. Restricting Proposition 4.7 to centers satisfying the criterion is likewise available, at a cost in generality that is the author's call.

The observation that multiplicity-one blocks contribute nothing, flagged in the second version of this memo on the strength of Dubrovin's zero-diagonal fact in canonical coordinates, is no longer a flag: Theorem 13 proves it in this normalization, from self-adjointness of $E\star$ and anti-self-adjointness of $\mu$ for the Poincaré pairing. So every route above needs to control only the coalesced blocks.

A separate scope reduction stands on its own, unchanged: a direct quantum Künneth computation for $X \times \mathbf{P}^1$ together with a direct computation for $\mathbf{P}^4$ would leave the blowup step as the only operation formula the headline theorem needs, with the genus-eight corollary waiting. Note that this cannot be had by triviality of the bundle alone — see the end of check one.

## 11 Placement in the manuscript

Stated by label because these insertions renumber Section 4. Items 1 to 3 and item 5 are unconditional and can be written now; the checks of Section 5 also make item 6 partly unconditional, since the comparison isomorphisms themselves are $z$-polynomial and the transport lemma applies to them with no receiver. With Section 7 the step that removes the positive-filtration bulk parameter is unconditional as well for the cubic ambient summand, so what is left conditional is the corresponding step at an arbitrary center. A ninth item is therefore owed: the rigidity theorem itself, as a lemma before `prop:framed-operations`, stated for the block type it covers and with its two hypotheses displayed rather than assumed. Nothing here should be written into the manuscript before the sign conventions are rechecked against the draft's normalization.

1. The ground field and solution algebra go immediately before "Apply Levelt–Turrittin after a ramification $z = w^e$", which already refers to a universal formal solution algebra that is never constructed.

2. Lemma 4 and Proposition 6 go after `def:framed-sixth-multiplicity`, converting that definition into a characterization, with Remark 5 stating the hypothesis that makes it available.

3. Theorem 7 becomes `lem:frame-transport`, after the paragraph beginning "The frame matters" and before `def:pro-laurent-gauge-group`.

4. In `lem:pv-base-change`, the asserted final clause and the corresponding sentence of its proof become a reference to `lem:frame-transport` — but only once the receiver repair makes that lemma applicable there, since that lemma lives over the pro-Laurent receiver.

5. At (4.1c) in `lem:formal-base-shift`, one sentence identifying $M^{\mathrm{bulk}}$ as the transported framed automorphism, in the module frame, with the solution-frame statement cited.

6. `prop:framed-operations` cites the lemma at its three load-bearing steps and says which field plays the role of $\mathcal{H}$ at each. Two of those steps are now unconditional: the transport across $\Psi$ and across $\Phi$ needs only that both are $z$-polynomial with inverses in the same ring, which their statements give, so the field there is $F(\!(z)\!)$ over the Laurent coefficient field and no criterion is verified. The step that removes the positive-filtration bulk parameter is the one that waits.

7. A sentence in `prop:framed-operations` fixing the normalization of the coordinate change, citing Iritani–Koto (5.11)–(5.12) and Iritani's Section 5.8.1, and noting that any residual

constant in a logarithmic Novikov direction is a divisor substitution already covered by (4.1). This is owed regardless of which repair lands, and is check two of Section 5.

8. `lem:divisor-tagging` carries the same obstruction — it embeds the gauge of `lem:formal-base-shift`, whose negative $z$-order is unbounded, together with every transported operator, into one ordered Hahn field — and needs whatever repair the receiver gets.

Mechanically, a new theorem-like environment must carry exactly one semantic label and a matching row in the claim map consumed by the manuscript's source-only gate, with an empty declaration list exactly when its coverage is recorded as absent, and the coverage snapshot string in both verification READMEs must be regenerated.

# 12 Corrections to the earlier versions of this memo

### From the fifth version to this one

Nothing in the fifth version is retracted; one of its hypotheses is discharged and one of its expectations is corrected.

1. Standing hypothesis (H2) of Section 7 is deleted. It is a theorem: Lemma 19 shows the decoupling gauge is an isometry of the Poincaré pairing, and Theorem 20 then forces $f \equiv 0$ pointwise on the germ, for every rank-two coalesced block of every smooth projective target. Theorem 21 and Corollary 24 now hold under (H1) alone. The flatness route to $f \equiv 0$ is kept as a remark, since it is what survives without the Frobenius property.
2. The shorter-endpoint-route subsection recorded the graded minimal polynomial as $S^5 = [3]$. On the numerical Grothendieck group that is incompatible with the count: it forces order ten, where the Serre operator computed from Riemann–Roch has characteristic polynomial $\Phi_6$ and order six, matching Kuznetsov's $S^3 = [5]$. The subsection now uses the verified relation, and states the surface obligation as (iii) with its two reductions and the non-minimal-surface counterexample to the weakened form. It also now proves (iii) outright by a minimal-model induction (Proposition 27), on Iritani's blowup decomposition read at the source plus two statements this route already uses; the endpoint route's surface exclusion is therefore no longer an open problem of its own.
3. The scope remark expected the rank-two mechanism to extend to a Jordan block of any size. It half does. Proposition 28 carries the commutant and no-splitting steps to size $m$, but Proposition 29 shows duality removes only the deepest of the pole coefficients the general shearing produces, so rank two is special rather than typical, and for $m \geq 3$ the base point must be settled by the grading, whose missing step is named there.

### From the fourth version to the fifth

Nothing in the fourth version is retracted. Section 7 is added, and three of its consequences change earlier readings.

1. Section 6 said the coalesced case was not reduced and named the block-splitting risk as the residual problem. Both are now settled for a rank-two Jordan block: Theorem 18 excludes

the splitting and Theorem 23 freezes the exponents. The section is left standing because its diagnosis — that the compressed pair is the wrong object — is what the shearing acts on.

2. The residual gap of Section 5 is closed for the cubic ambient summand by Corollary 24. It remains open as stated for arbitrary smooth projective $T$, which is a statement about arbitrary blocks, not about transport.

3. Cai's Proposition 6 already asserts the $\pm 1/6$ exponents for the big quantum connection, by a bulk gauge of exactly the pro-Laurent shape this memo scrutinized. That argument is sound with the bulk kept formal; the receiver discussion was always about specialization, not about the formal statement. Section 7 makes the point moot by producing constants.

### From the third version to the fourth

Both are in Section 6, and both are at the coalesced blocks, which are the ones that matter.

1. The off-block projector equation was derived as $(\partial_a U)_{ji} + (u_j - u_i + N_j)X_{ji} = 0$, dropping the term $-X_{ji}N_i$. The correct equation is $\mathcal{S}_{ji}(X_{ji}) = -(\partial_a U)_{ji}$ with $\mathcal{S}_{ji}(X) = (u_j - u_i)X + N_j X - X N_i$. The closed form $D_a \mu_i = [C_{a,i}, S_i]$ therefore holds only when the block itself is semisimple, and the blocks carrying $\nu_6$ are exactly the ones where it is not; in the draft's cubic block $J_0 \neq 0$.

2. The block was modelled as $z\partial_z - z^{-1}U_i + \mu_i$. After formal decoupling a nonsemisimple block carries higher powers of $z$, and its exponents depend on them: in the cubic block the $z^2$ coefficient's entry $-8/81$ enters $\rho^2 + \rho + 5/36 = 0$, and deleting it moves the exponents from $\pm 1/6$ to $\pm 1/18$. So the isomonodromy criterion stated there is a statement about semisimple blocks and is not a reduction of the residual problem. It is restated with that hypothesis.

The simple-block theorem survives, with one wording repair: the formal solution of a simple block is $e^{-u_i/z}$ times a single-valued formal unit, not times a constant, since decoupling can generate higher $z$-terms there too. The monodromy conclusion is unaffected.

### From the second version to the third

Both changes come from running the three checks against the sources.

1. The framing route of Section 5 was called the probable repair. It is not a repair at all: its governing theorem lives at the large-radius limit point; its blowup case, though stated with existence in Theorem 5.22 and uniqueness in Theorem 5.24, is not derived from Theorem 4.34, whose hypotheses do not cover the unequal block sizes, and its existence is referred to the earlier decomposition work; and the gauge it supplies lands at the shifted base point the bulk gauge exists to undo.

2. The second version wrote that the unbounded negative loop order is the draft's own choice rather than something the comparison theorems impose. That is wrong, and it was the ground for the recommendation. Iritani–Koto transport along the same object, their fundamental solution $M$ of Section 5.8, with the same per-bulk-degree $z^{-1}$-polynomiality the draft derives in (4.1a).

Two findings were gained in exchange, and they narrow the problem rather than restate it: the comparison isomorphisms need no receiver, and the residual gap is one displacement statement. Both are in Section 5.

### From the first version to the second

For readers of the first version, these are the substantive changes.

1. The Constants Lemma was stated without the constant-coefficient hypothesis and is false without it; see Remark 5. The framed-operator characterization inherits the hypothesis.
2. The scissors were only implicit. The transport theorem has no verified instance in the application, and that now appears where the theorem does.
3. The claim that the criterion is scale-invariant, and the inference that the separating parameter $L$ cannot be tuned, were both wrong. Tuning $L$ down is now the first proposed route.
4. The cubic-endpoint bullet compared two different normalizations. Corrected above.
5. The finite-tensor objection to the receiver was wrong twice over: the pro-Laurent gauge lies in the Hahn factor alone, and the splitting gauge's coefficients lie in one finite extension of $H_{0,j}$. It is replaced by the gauge-relation argument.
6. The remark that a tensor product of two fields is not a domain was asserted as a general principle, which is false. The correct statement is that $\mathscr{R}_j$ need not be one.
7. The obstruction is now given in its order-free form, which answers two of the questions the first version left open: no completion helps, and the criterion is not an artifact of the ordered receiver.
8. Smaller repairs: the freeness proof (the relevant quotient group has torsion), the commutation of $\sigma$ with the derivation, the inverse of $M$, the range of the exponents, the ramification factor in the splitting rate, and the divisor-tagging observation, which is new.

## 13 Questions

The first question of the second version — whether the two connections compared in Proposition 4.7 lie in the class Theorem 4.34 governs — has been answered no, three times over, in Section 5. Questions 1 and 2 of the fourth version are answered in Section 7: the object to deform is the residue of the sheared block, which sees the $z^2$ coefficient by construction, and its characteristic polynomial is constant; and the displacement cannot split the block, because $\det N$ satisfies a linear equation with zero initial value. Question 3 is no longer on the critical path, since the coalescence literature is not used. These remain.

1. Does the argument of Section 7 extend to a single Jordan block of arbitrary size, and to a semisimple block once the resonance $\rho_i - \rho_j = -1$ is excluded? This is what the draft's own presentation needs at every intermediate fourfold. The endpoint no longer waits on it, by Section 8, so this is now a question about how the result is presented and about the general

stabilization statement, not about whether the endpoint holds. Partially answered: the commutant and no-splitting steps extend to any Jordan size (Proposition 28), but duality kills only the deepest pole coefficient (Proposition 29), so for $m \geq 3$ the base-point input must come from the grading instead. The precise remaining step is $\mu$-equivariance of the decoupling gauge on the zero block.

2. *Answered, and the hypothesis is gone.* Hypothesis (H2) — that the sheared block is regular singular at the base point — was structural, not computed. What was missing, the same statement for the decoupling correction, is Lemma 19: the decoupling gauge is an isometry of the Poincaré pairing, so the parity rule (8.5) applies to the corrected coefficients, and Theorem 20 then gives the vanishing pointwise on the germ for every smooth projective target.

3. Does the factor $M' = \mathrm{id} + O(z^{-1})$ of the sources' Birkhoff factorization preserve the primitive-sixth multiplicity? Now a cross-check rather than a route.

4. For the low-dimensional centers, can primitive-sixth vanishing be proved pointwise in the coordinates Iritani's specialization produces, given that the nef-canonical case already yields an integral-$z$ gauge to a regular-singular connection and that Theorem 13 disposes of every multiplicity-one block? A pointwise statement there needs no transport at all, which is the shape the endpoint theorem wants.

5. Which presentation should the manuscript take: the operation formulas with $\nu_6$ transported summand by summand, which needs the generalization in item 1, or the atom formulation, in which the intermediate fourfolds never appear because the weak-factorization bookkeeping happens at the level of atoms as objects?